Polar Organometallic Reagents: an Evergreen Tool for Tuning the Reactivity of Unsaturated Systems

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Abstract: The advice of professor Dieter Seebach "*Thinking of polar organometallics compounds as carbanions is an impoverishment rather that a simplification*" (ISCC, Durham 1984), perfectly fits the statement of professor Manfred Schlosser, who coined the term *polar organometallic*", according which no difference in reactivity patterns of organometallic reagents can be rationalized without taking into account the metal counterion and its specific interactions with the accompanying carbon backbone. Both statements stress the need of considering the metal as a key component of these reagents, that can initiate and govern the subsequent reactions. The class of the so-called *polar organometallics* comprises the reagents of the most common alkali and alkaline earth metals and extend into the area of organozinc derivatives. All these reagents are characterized by a polar carbon-metal bond and the pattern of polarity vary significantly with the metal.

Metalation is one of the widely used synthetic routes in order to obtain a labile and reactive carbon-metal bond from the relatively inert carbon-hydrogen bond. Until recently, alkyllithium reagents and bulky lithium amides have been commonly employed for this purpose. More recently, some groups have pioneered alternative metalating agents made up of two or more distinct compound types. Several examples include Schlosser's superbases, Kondo and Uchiyama's 2,2,6,6-tetramethylpiperidide (TMP)-zincate complexes, and Knochel's turbo-Grignard reagents.

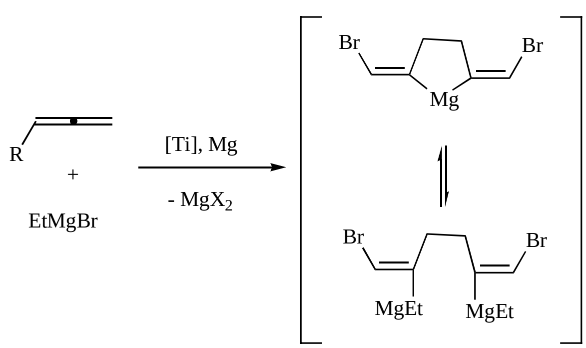
In this review, we wish to report the most recent contributions appeared in the literature from 2005 to date, concerning the use of polar organometallics in synthesis. The topic has been divided into three sections, the first of which deals with metalated 1,2-dienes and the second one with metalated 1,3-dienes (the metal is in turn lithium, potassium or magnesium). The third section is dedicated to miscellaneous applications that cannot be comprised into the previous classification.

Key Words: Polar organometallics, dienes, metalated dienes, mixed bases.

1,2-DIENES

1,2-Dienes and Magnesium

It is reported in the literature that, apart from the hydrocarbon halide, the presence of functional groups influences the selectivity of OMC (organomagnesium compounds) formation. The introduction of metal complex catalysis allows the scope of Grignard reagents applicability to be expanded and to elaborate promising procedures for practical utilizations based on hydro- carbo- and cyclomagnesiation reactions of unsaturated compounds. Recently a new intermolecular cyclomagnesiation of terminal allenes has been proposed by Dzhemilev and coworkers. This was accomplished with the aid of Grignard reagents in the presence of Cp_2TiCl_2 as catalyst.



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Scheme 1. Cyclomagnesiation of terminal allenes.

As illustrated in Scheme **1**, the treatment of terminal allenes, with Grignard reagents in the presence of chemically activated magnesium and catalytic Cp_2TiCl_2, allows the achievement of 2,5-dialkylidenemagnesacyclopentanes **1** in quantitative yields.

When this method was applied to α, ω-diallenes in THF at 20 °C, it led to a stereoselective intermolecular cyclomagnesation simultaneously of two or more molecules of the initial diallene **2**. Moreover it provided macrocyclic organomagnesium compounds (**3**) which brought magnesacyclopentane fragments **4**. They could afford unsaturated hydrocarbon macrorings, if hydrolyzed (Scheme **2**) [1].

The reaction was then extended to cyclonona-1,2-diene that produced 2-magnesatricyclo$[10.7.^{1,12}0^{3,11}]$nonadeca-3(4),19-diene **5** [2]. Acid hydrolysis and deuterolysis of OMC led to compounds **6** and **7** with yield higher than 85%. The reactions were carried out in Et_2O, for 4 h, using a 5% of catalyst (Scheme **3**).

Ma and coworkers have recently optimized a controllable and stereoselective α-acylation and -allylation reaction of the magnesium dienolate intermediates **8**. They were generated *in situ* from the Fe(III)-catalyzed reaction between 2,3-allenoates and Grignard reagents with different electrophiles. 2-Acylated or -allylated 3(*Z*)- or (*E*)- (**9** and **10**) alkenoates were obtained in dependence on the nature of the electrophiles and reaction conditions (Scheme **4**) [3].

The authors hypothesized that the distinct stereoselectivity might be caused by the isomerization of metallic *Z*-1,3-dienoate and *E*-1,3-dienoate via the intermediacy of *anti*-allylic MgCl and *syn*-metallic species.

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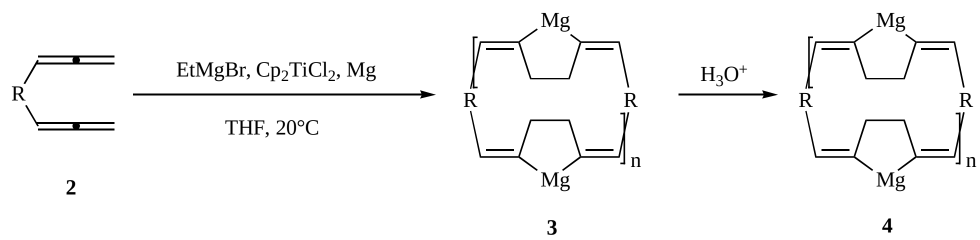
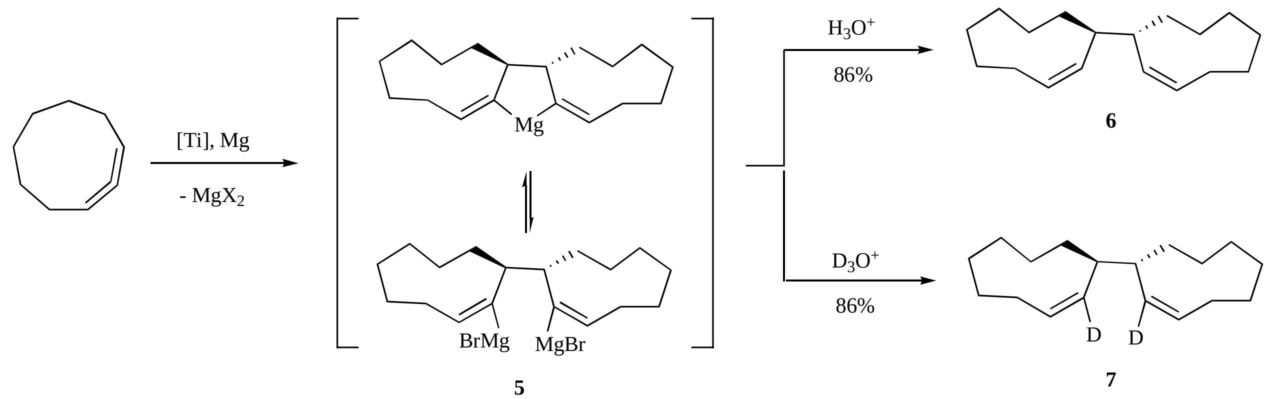
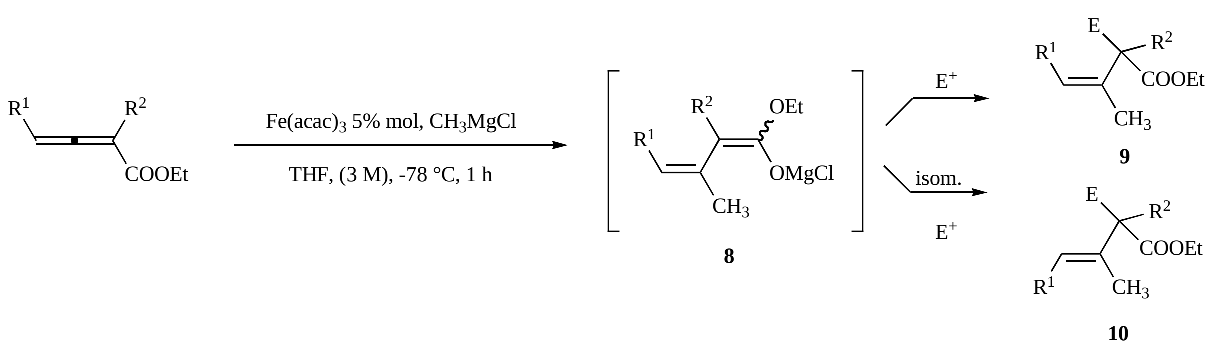


Scheme 2. Cyclomagnesiation of α,ω-diallenes.



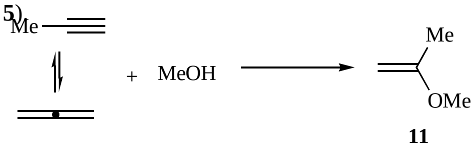
Scheme 3. Intermolecular Cyclomagnesiation of cyclonona-1,2-diene.



Scheme 4. Stereoselective α-acylation and -allylation reaction of the magnesium dienolate intermediates.

1,2-Dienes and Potassium

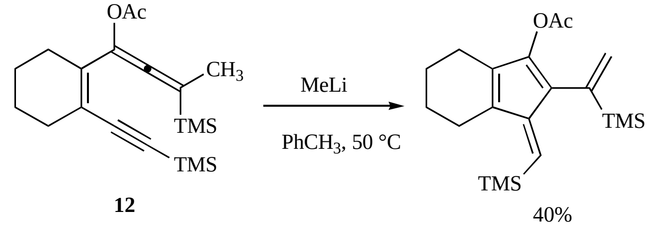
A new synthesis of 2-methoxypropene (**11**) by the direct isopropenylation of methanol with a propene-allene system in superbasic catalytic system was reported [4]. It is well known that this reaction is normally carried out in strong reaction conditions, as for example 28–38 Torr and 140–190 °C (catalyst, NaOH, 40% yield) or in a gas phase at high temperatures (180–300 °C) always accompanied by 2,2-dimethoxypropane as a byproduct. In this case it was accomplished in *tert*-BuOK/*N*-methylpyrrolidone (NMP), *tert*-AmOCs/NMP and KOH/DMSO systems where the methanol readily reacted with the propyne-allene mixture under atmospheric mixture at 100–200 °C to afford 2-methoxypropene **11** chemo- and regioselectively in quantitative yield and complete conversion (Scheme **5**).



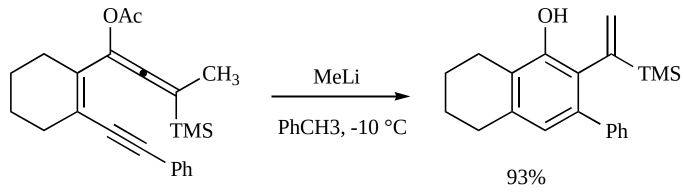
Scheme 5. Synthesis of 2-methoxypropene in superbasic conditions.

1,2-Dienes and Lithium

A series of acetoxy-substituted enyne-allenes, fused to cyclopentene and cyclohexene ring system, were synthesized and treated with methyllithium to generate the corresponding enolates. In comparison to the previous results reported in literature, where only the C^2–C^6 cycloaromatization were reported, in this case both the thermal C^2–C^6 cyclization of the bis-TMS system **12** (Scheme **6**) and the cryogenic C^2–C^7 cyclization of the mixed TMS/phenyl-substituted allene/enyne **13** were observed (Scheme **7**) [5].



Scheme 6. Thermal C^2–C^6 cycloaromatization of cycloannulated allene-enyne.



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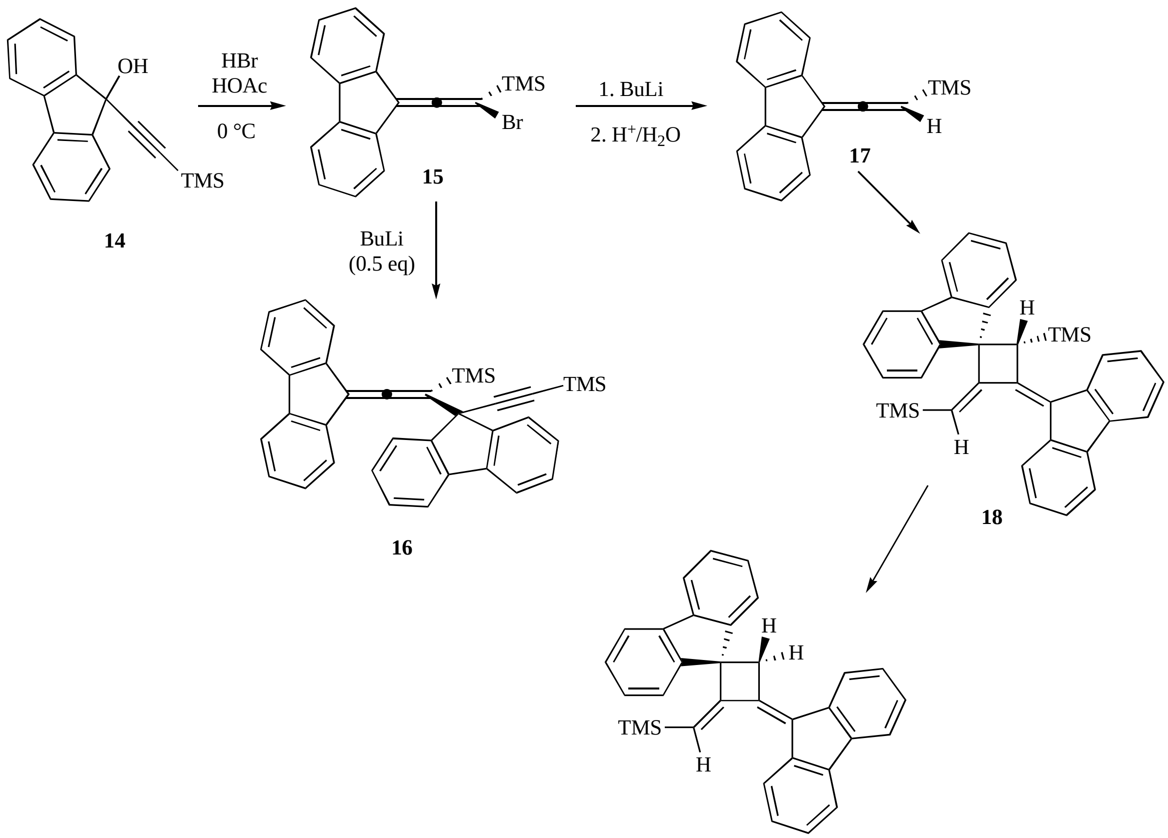
Scheme 7. Oxanion promoted C^2–C^7 cycloaromatization of cycloannulated allene-enyne.



14 **15** **17** **16** **18**

Scheme 8. Bromo and silyl-allenes and their corresponding dimmers.

The authors hypothesized that in the last case, the anomalous cyclization of **13** could be explained on the basis of the greater steric demand of the TMS substituent. Moreover the cyclization of oxyanion-substituted enyne-allenes in this study occurred at far lower temperatures than the similar process of neutral enyne-allenes. The C^2–C^7 cyclization of **13** here described, represents one of the fastest examples of a Myers-Saito cycloaromatization ever reported.

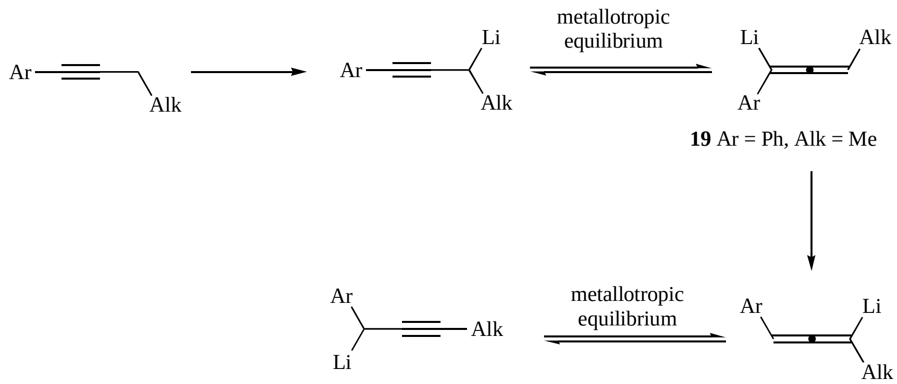
In continuation of their studies on the syntheses and dimerizations of fluorenylidene-derived allenes to form bis(alkylidene)cyclobutanes and tetracenes, McGlinchey and coworkers proposed a new synthetic methodology concerning substrates which incorporated silicon and/or halogen. Since tetracenes are known to be electroluminescent, with the aim of controlling the luminescent and other photophysical or other electro-conductive properties by introduction of a wider range of functional groups onto the precursor allenes [6].

The desired allenes **15** was obtained from the corresponding substituted ethynyl-9*H*-fluorene **14** by treatment with dilute HBr. The treatment of the diene **15** with a deficit of BuLi, produced the dimer **16**. In contrast the addition of **15** to butyllithium and subsequent hydrolysis, yielded the desired 3,3-(biphenyl-2,2'-diyl)-1-(trimethylsilyl)allene **17**. However when the allene **17** was stirred in cyclohexane for 24 h, it gradually formed the head to tail dimer **18** (Scheme **8**).

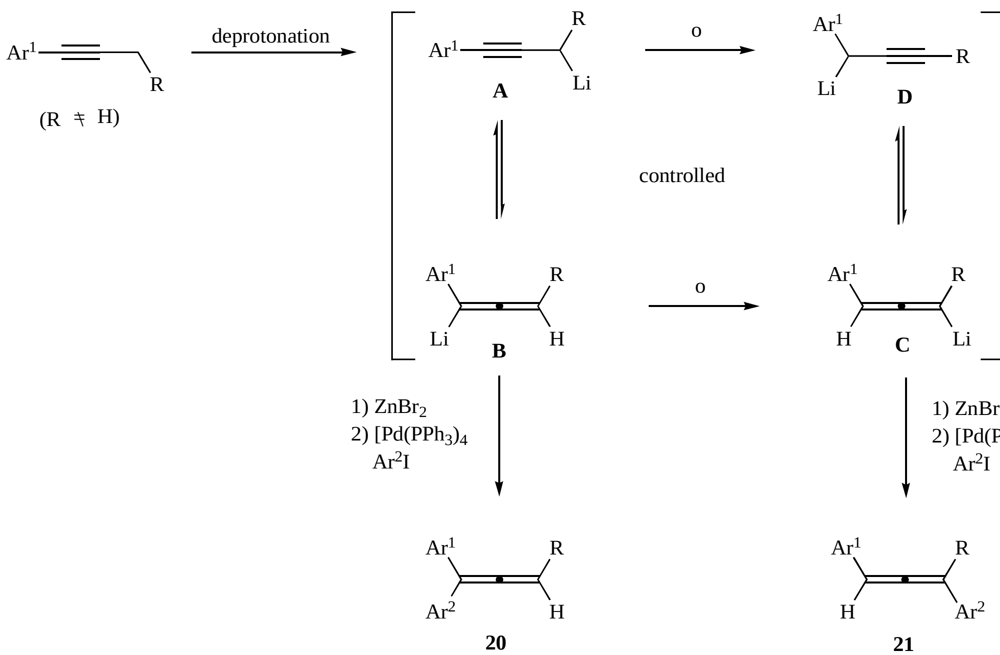
The same authors studied the reactivity of an allenylphosphine in the same conditions and synthesized the corresponding chromium, iron, ruthenium and gold complexes, versatile ligands [7].

The 1,3-Li/H shift of 1-Aryl-1,2-alkadienyl (**19**) reagents were studied from an experimental and theoretical point of view [8]. A new mechanism, which involved the exchange of equilibrated protons initiated by a catalytic amount of a proton donor, was proposed. The most efficient donor appeared to be the diisopropylamine. Surprisingly, the use of drastic reaction conditions, with the aim to protect the reaction medium from air and water, forbade the formation by transposition of the desired product. As long as the mechanism remained uncertain, the 1,3-shift was limited to the alkyl/phenyl substituents (Scheme **9**).

Four protocols to control the 1,3-lithium shift were developed by Ma and coworkers [9]. Highly selective formation of 1,1-diaryl allenes (**20**) or 1,3-diaryl allenes (**21**), by Pd-catalyzed Negishi



Scheme 9. 1,3-Li/H shift of 1-Aryl-1,2-alkadienyl reagents.

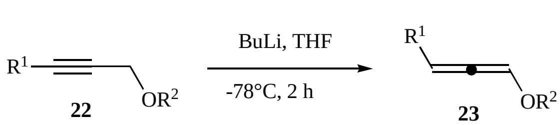


Scheme 10. A controlled 1,3-lithium shift reaction.

coupling, was realized with the application of different reaction conditions. Since the starting compounds are easily available and structurally different, the proposed methodology might be applied in organic synthesis. Moreover experimental studies and DFT calculations suggested that LDA acts as a carrier to transfer the proton intra/intermolecularly with 1-phenylbuta-1,2-diene as intermediate. Furthermore the isomerization step was accelerated by THF solvation (Scheme **10**).

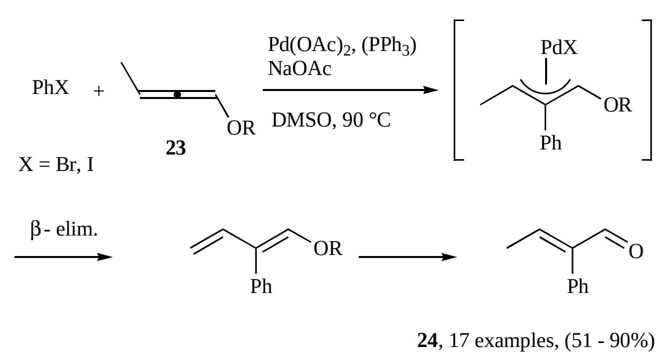
Lithium Alkoxy 1,2-Dienes

The isomerization of protected alkynols (**22**) was exploited to produce the protected 1,2-dienols (**23**) in our laboratory (Scheme **11**). The reaction was accomplished using 2 eq of BuLi, THF as the solvent, at –78 °C. As reported in literature in the case of internal alkynes, a mixture of propargyl and allenyl ethers was obtained [10].



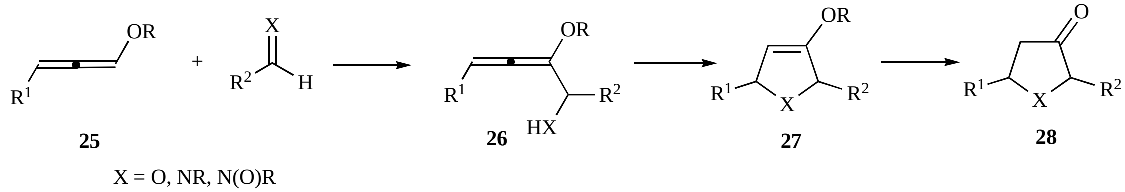
Scheme 11. Isomerization reaction of alkynes to allenes in the presence of BuLi.

The so obtained alkoxyallenes were used for the regio- and stereoselective preparation of α-arylated α,β-unsaturated aldehydes **24**, as illustrated in Scheme **12**.

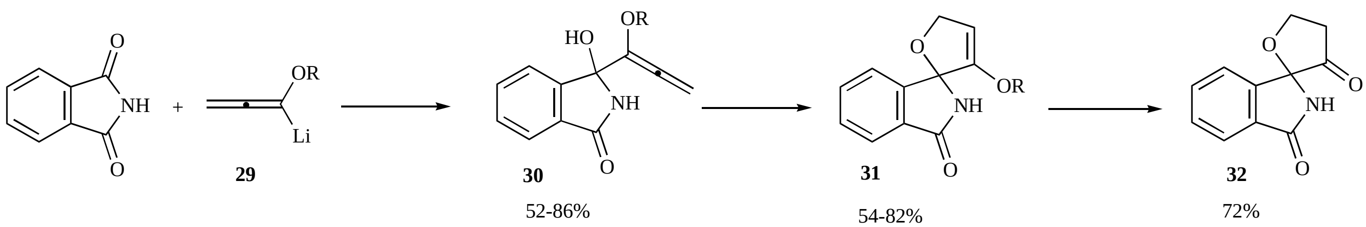


Scheme 12. Heck coupling of aryl halides and allenols to afford α-arylated α,β-unsaturated aldehydes.

During the last years the Reissig' s group gave a strong contribution to the exploitation of lithiated alkoxyallenes as useful functionalized C-3 units in the construction of various classes of heterocycles, natural products or carbohydrate mimetics.



Scheme 13. General reactivity of methoxyallenes with electrophiles.

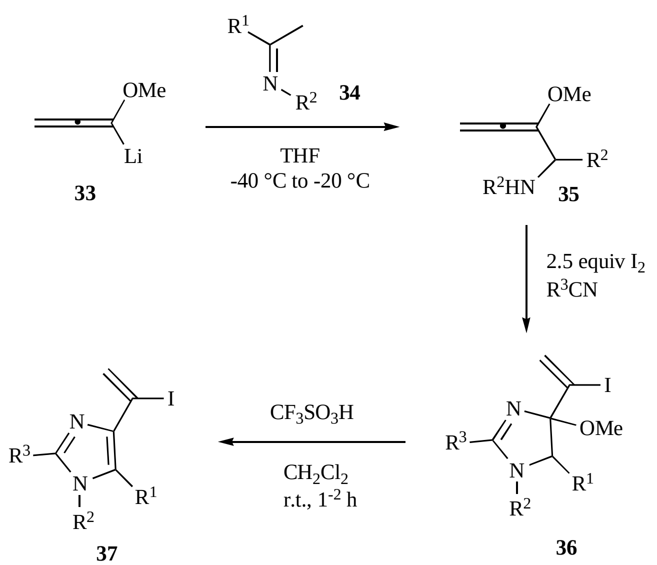


Scheme 14. Reaction of alkoxyallenyl lithium with phthalimide.

As represented in Scheme **13** the lithiation of methoxyallenes **25** with *n*-BuLi and subsequent reaction with carbonyl compounds, imines or nitrones lead to the corresponding adducts **26**. Their cyclization affords five- or six- membered heterocycles **27** and **28** such as furanes, pyrroles and 1,2-oxazine derivatives.

An excess of the lithiated alkoxyallenes **29** react with phthalimide to afford primary allene adducts **30** that could in principle lead through a ring-closing reaction to either furan or pyrrole derivatives. Specifically, treating **30** with *tert*-BuOK in DMSO spiro compounds **31** were obtained in acceptable yields (Scheme **14**). Acidic hydrolysis of **31** furnished dihydrofuran-3-one derivative **32** [11].

The addition of lithiated methoxyallene **33** (generated from methoxyallene and *n*-BuLi) to imines **34** provided allenyl amines **35** [12], which upon reaction with iodine in various nitriles as solvent furnished diastereomeric mixtures of dihydroimidazole derivatives **36**. Strong acids such as trifluoromethansulfonic acid promoted the elimination of CH_3OH to afford 1-iodovinyl-substituted imidazoles **37** (Scheme **15**).



Scheme 15. Synthesis of 1-iodovinylimidazole derivatives **37** from methoxyallene, imines, iodine and nitriles.

Due to the presence of the iodovinyl functional group, derivatives **37** can be further transformed into different higher substituted and functionalized heterocyles by means of traditional transition metal catalyzed cross-coupling reactions.

An interesting approach to the synthesis of Codonopsinine has been proposed on the base of the proposed methodology [13].

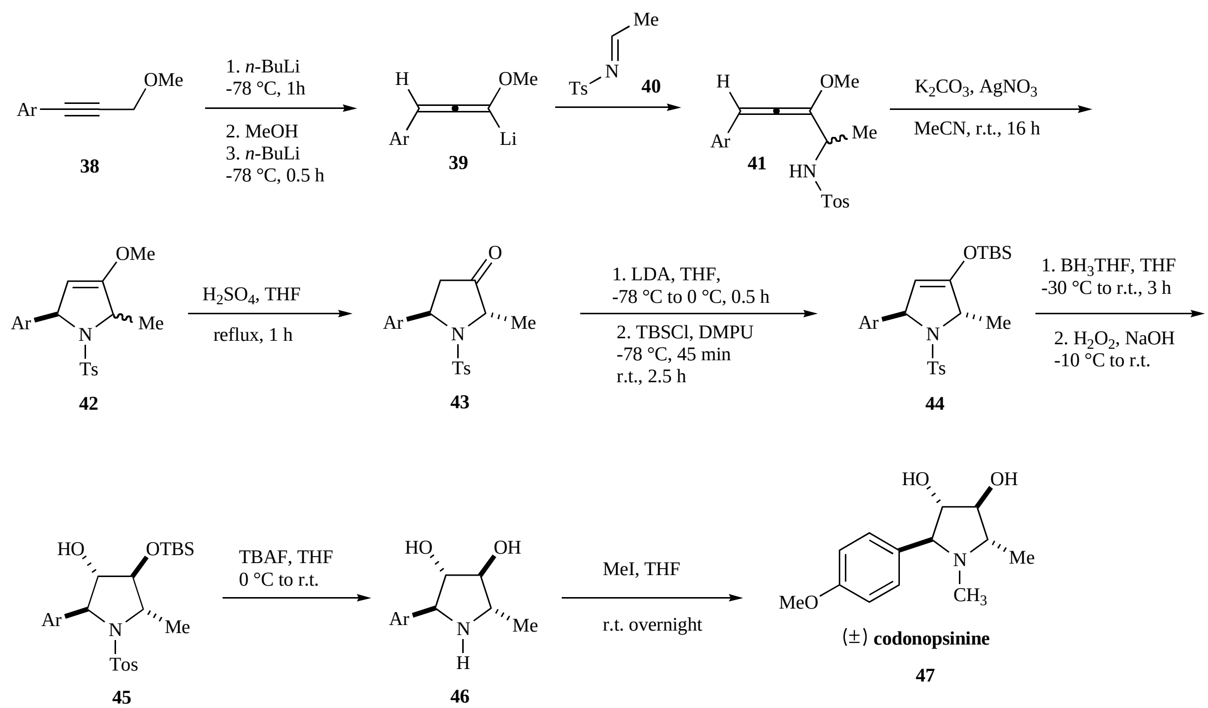
Compound **42** (Scheme **16**) was obtained resorting the usual procedure in 78% yield as a mixture of diastereoisomers that have been separated by fractional crystallization. The following steps are represented only for the *trans* diasteroisomer, but the same pathway has been successfully applied to the *cis* under the same reaction conditions. So, acid-promoted hydrolysis of the enol ether moiety of **42** gave ketones **43** which was converted into silyl enol ethers **44** under standard conditions. Hydroboration of the silyl enol ether moiety with borane in THF installed the second hydroxyl group in **45** with the expected stereochemistry. After desilylation with TBAF and detosylation of the nitrogen, the tetrasubstituted pyrrolidine derivative **46** was obtained in very good yield. Finally *N*-methylation was smoothly achieved by stirring **46** with methyl iodide in THF at room temperature to obtain the expected (t-3, c-4, t-5)-**47** (±)-codonopsinine. The same synthetic sequence conducted on **42** *cis* led to the epimer of this alkaloid.

The rare deoxy sugar L-cymarose **57** (Scheme **17**) has been prepared starting from dihydrofuran **49** this in turn obtained from the reaction of methoxyallene **33** with *O*-trityl protected 2-hydroxy propanal **48** [14]. The DDQ mediated oxidation of **49** efficiently leads to dicarbonyl compound **50.** The trityl group was then removed and concomitant cyclization cleanly led to isopropyl acetal **51.** Hydrogenation of **51** was achieved with 10 mol % rhodium on aluminum oxide and 1 bar of hydrogen pressure and led to the isolation of diastereomeric glycosides **52** and **53** in 67% combined yield.

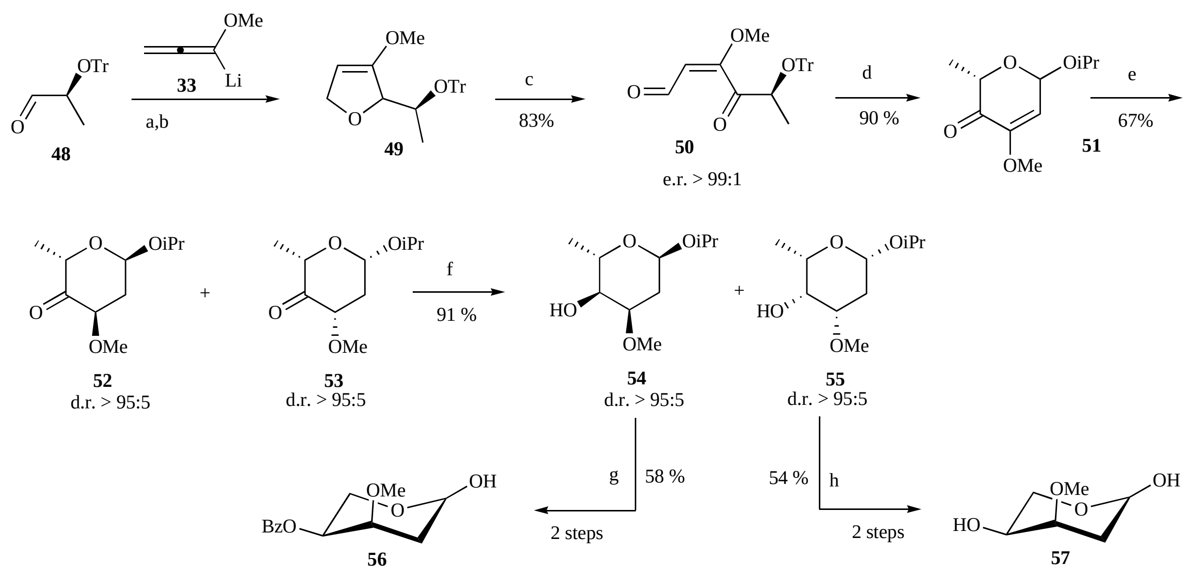
The mixture of ketones **52** and **53** has been then reduced with L-selectride with perfect stereocontrol to give alcohols **54** and **55**. At this stage the diastereoisomers could be readily separated by column chromatography. Direct hydrolysis of **55** yielded the free L-cymarose **57**.

Methoxyallenes **33** have been added to naphthaldehyede derivatives **58** (Scheme **18**) thus obtaining useful intermediates for the synthesis of members of the rubromycin family (Fig. **(1)**) [15].

The synthesis of the key compound **58** can be achieved in a 12-steps sequence in an overall yield of 9%. This was then combined with lithiated methoxyallene **33** and the resulted allene adduct was subsequently hydrolyzed to the expected α-hydroxyenone in quantitative yield.

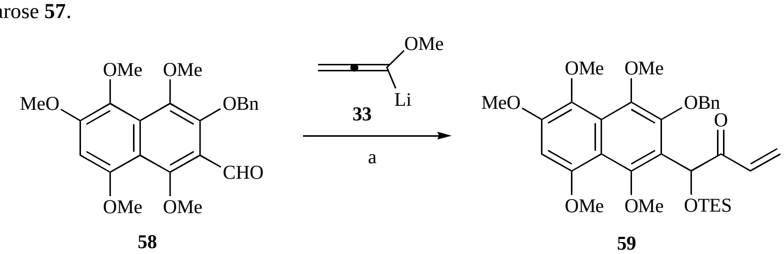


Scheme 16. Synthesis of Codonopsinine.



conditions: a) 3 equiv **9,** Et_2O, -78 °C; b) 0.5 equiv KOtBu, DMSO, 60 °C; c) 2 equiv DDQ, CH_2Cl_2/H_2O 20:1, RT; d) 0.3 equiv I_2, $HC(OiPr)_3$, $iPrOH/CH_2Cl_2$, 60 °C; e) 10 mol% Rh/Al_2O_2, 1 bar H_2, EtOAc, RT; f) $Li(sBu)_3BH$, THF, -78 °C; g) BzCl, DMAP, pyridine, CH_2Cl_2, RT; h) 2 N aq. HCl, THF, RT, Bz = Benzoyl.

Scheme 17. Synthesis of L-cymarose **57**.



a: (i) **33**, nBuLi, THF, -40 °C, 1h, (ii) -78 °C

Scheme 18. Synthesis of a-triethylsilyloxy enone **59** from aldehyde **58** and methoxyallene **33**.

A similar synthetic approach has been successfully applied to the synthesis of (±)- Annularin H [16].

Commercially available methyl 3-oxovalerate **60** (Scheme **19**) was converted into protected aldehyde **61** within 3 steps. Addition

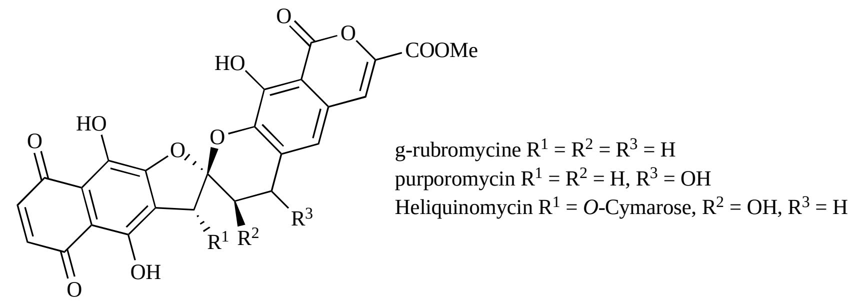
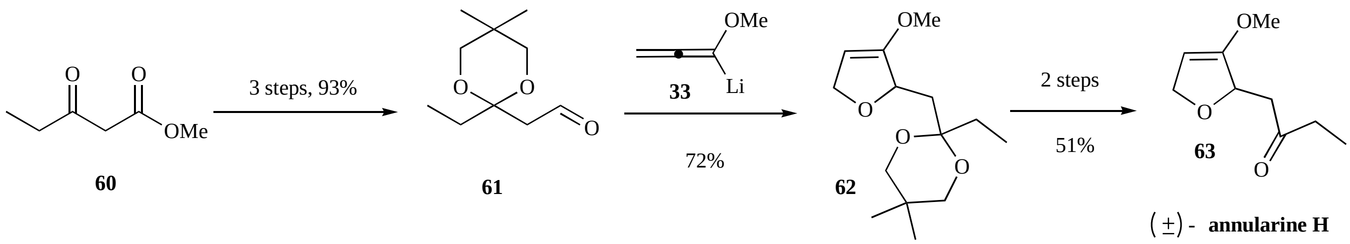
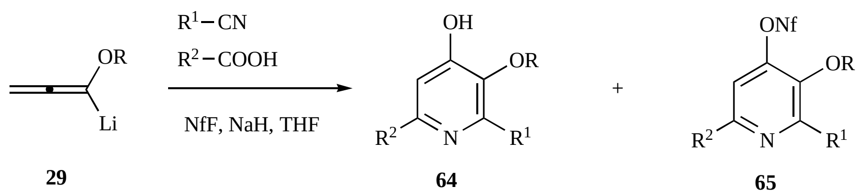


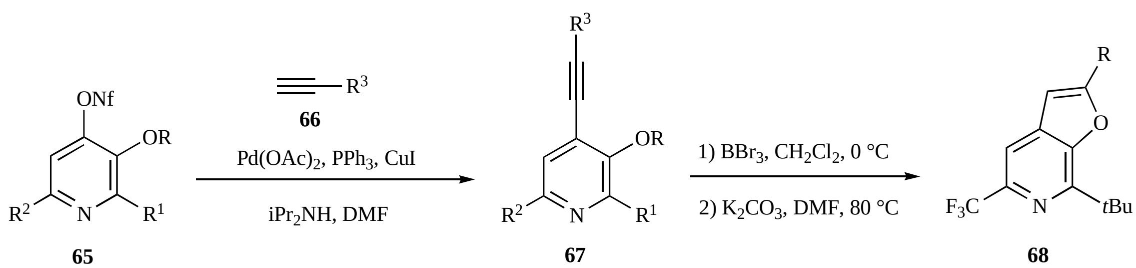
Fig. (1). Compounds of the rubromycine family.



Scheme 19. Preparation of (±)-annularin H.



Scheme 20. Three component synthesis of functionalized hydroxyl pyridines or nonaflates.



Scheme 21. Synthesis of 3-alkoxy-4-alkynylpyridines to furo[2,3-*c*]-pyridines.

of lithiated methoxyallene **33** and 5-*endo* cyclization with AuCl/pyridine provided dihydrofuran **62**, which upon allylic oxidation and deprotection furnished the racemic natural product **63** with good overall efficiency.

Functionalized pyridines derivatives have been synthesized in good yield by a three component reaction involving an alkoxyallene **29**, a carboxylic acid and a nitrile (Scheme **20**) [17].

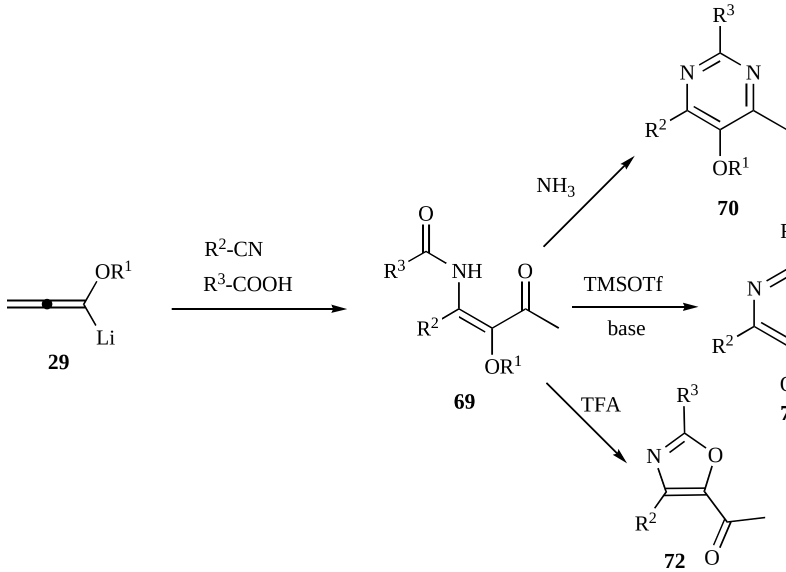
Lithiated alkoxyallenes are thus employed in a cascade reaction affording pyridines derivatives. Many substituents can be incorporated to the pyridine core at C-2 and C-6, moreover the oxygen function at C-4 allow the palladium catalyzed substitution which enhance the versatility of the proposed methodology. To this purpose nonaflates **65** can be directly obtained from the crude products by a simple protocol.

The nonaflates thus obtained could be coupled according to a Sonogashira reaction with several terminal alkynes to provide new 4-alkynyl-substituted 3-alkoxypyridine derivatives **67** (Scheme **21**) [18]. Subsequent treatment with boron tribromide and potassium carbonate led to furo[2,3-*c*]pyridine **68** in good yield.

The scope of this alkoxyallene-based pyridine synthesis was expanded to enantiopure carboxylic acids and nitriles as starting materials [19]. The enantiomeric purity of the chiral α–secondary carboxylic acids and nitriles was completely preserved during this reaction sequence thus allowing the one-pot preparation of a whole range of 4-hydroxypyridines or their 4-pyridinone tautomers in good yields.

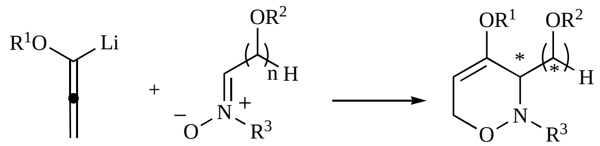
The intermediate enamides **69** arising from the three component reaction of alkoxymethyl lithium allene **29**, carboxylic acids and nitriles are ideal precursor for the synthesis of highly functionalized 4-methylpyrimidines **70** [20] and 4-hydroxypyridines derivatives **71** **[8]**using either ammonium salts or trimethylsilyl trifluoromethansulfonate and base (Scheme **22**) [21].

The cyclization of enamides **69** to the correspondent oxazoles **72** was carried out with an excess of trifluoroacetic acid acting as a solvent and reagent. The correspondent oxazoles were obtained in moderate to good yields (39–99%). The method allows a high flexible substitution pattern at C-2 and C-4 position. The acetyl group at C-5 can be further functionalized employing standard reactions.



Scheme 22. Synthesis of 4-methylpyrimidines **70**, 4-hydroxypyridines **71** and 5-acetyloxazoles **72** via alkoxysubstituted enamides **69**.

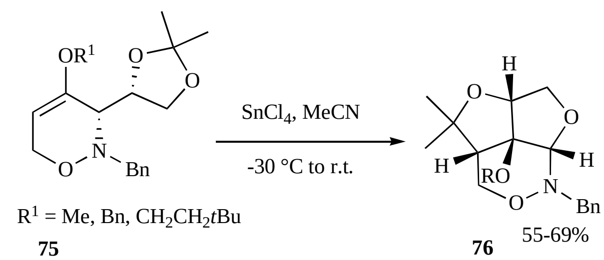
In the past years Reissig reported the synthesis of enantiomerically pure 3,6-dihydro-2*H*-1,2-oxazines **74** by means of the stereocontrolled addition of lithiated alkoxyallenes **29** to carbohydrate-derived nitrones **73** followed by spontaneous cyclization of the primary allene adducts (Scheme **23**) [22].



29 **73** **74**

Scheme 23. Synthesis of 3,6-dihydro-2*H*-1,2-oxazines from lithiated alkoxyallenes and nitrones.

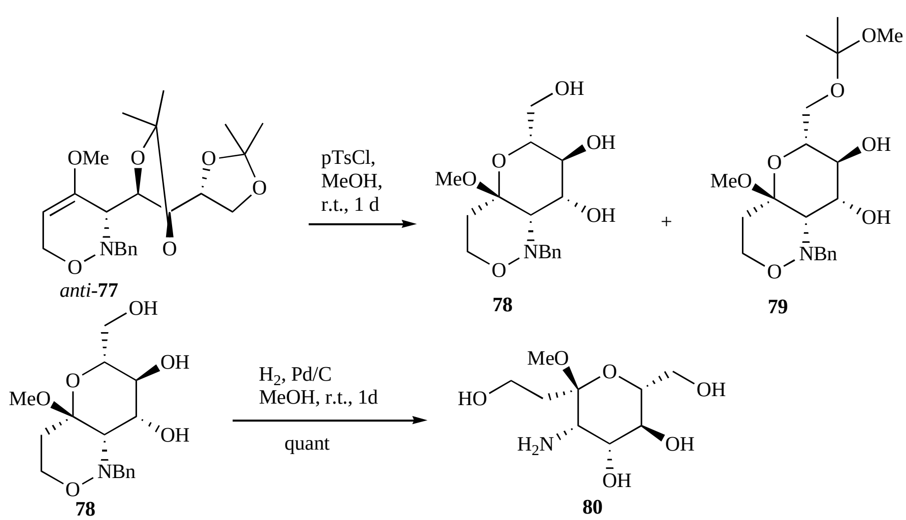
More recently, the behavior of 1,3-dioxolanyl-substituted 1,2-oxazines **75** *syn* or *anti*, prepared in a stereodivergent manner by [3+3]-cyclization of lithiated alkoxyallenes and D-glycerinaldehyde-derived nitrones in the presence of Lewis acid has been considered (Scheme **24**).



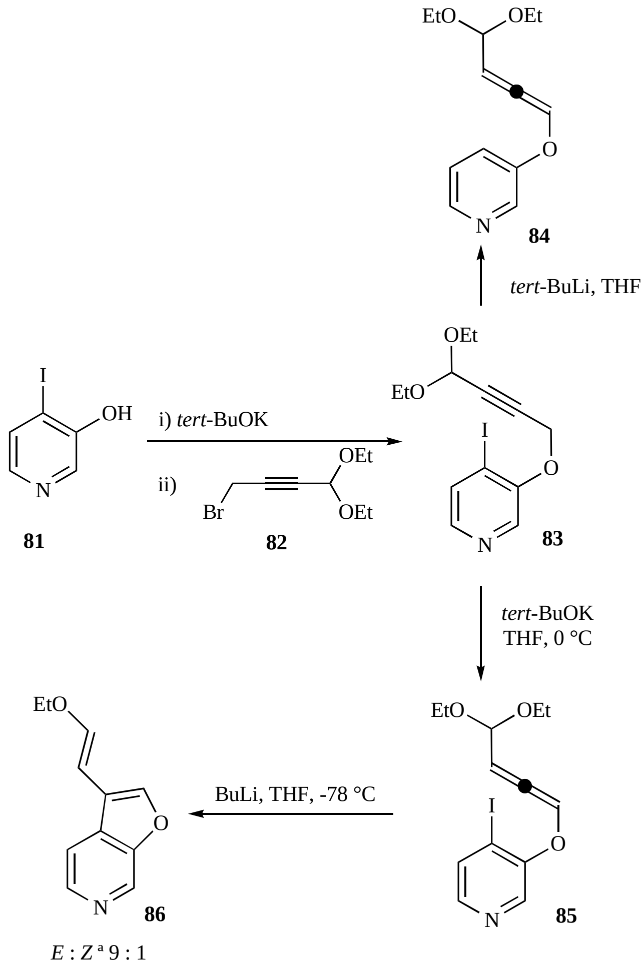
Scheme 24. Lewis acid promoted rearrangement of *syn*-**75**.

When 4-alkoxy-substituted-1,2-oxazines *syn* were exposed to tin tetrachloride products **76** with three annulated heterocycles were recovered in reasonable yields. The resulting tryciclic products **76** are good starting points for further transformations.

1,2-Oxazines of type **74** have been used as crucial intermediates in the synthesis of carbohydrate derivatives (Scheme **25**). D-arabinose-based 3,6-dihydro-2*H*-1,2-oxazine *anti*-**77** has been transformed by acid catalysis in methanol into the pyran-1,2-oxazine derivative **78** in 74% yield together with **79** in 7% yield.



Scheme 25. Synthesis of carbohydrate derivatives.



Scheme 26. Heterocyclization of propargylic and allenic acetals.

When pyrano-1,2-oxazine **78** was exposed to hydrogen in the presence of palladium on charcoal the expected deprotected amino carbohydrate **80** was quantitatively obtained [23].

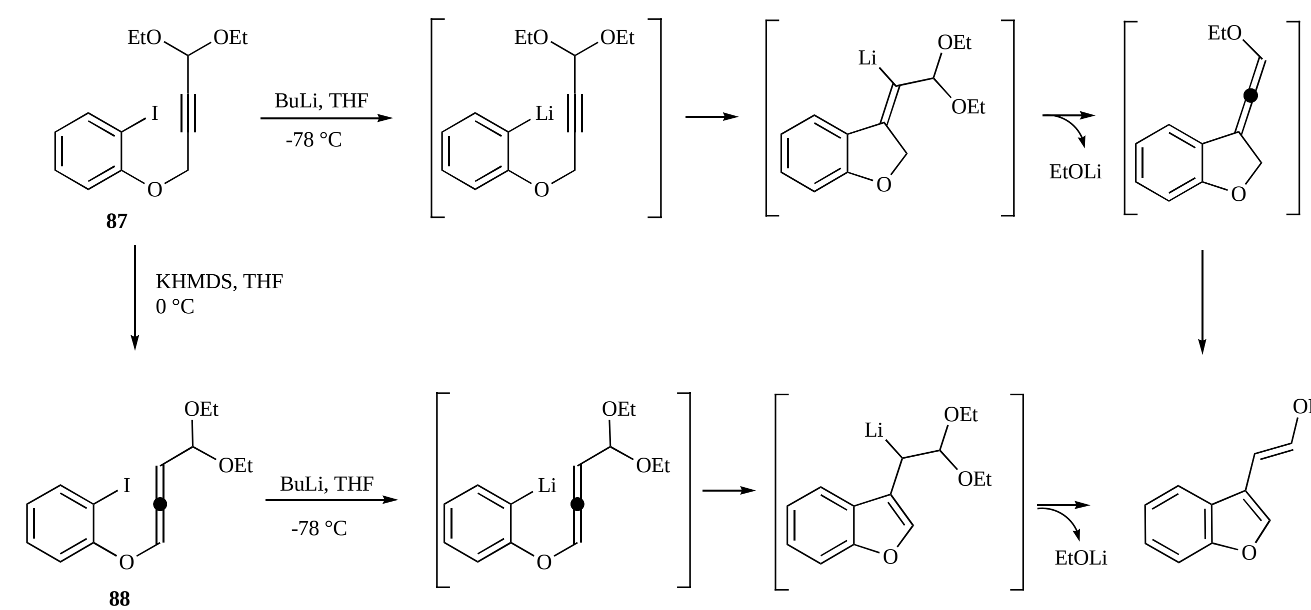
1,2-Dienes From Propargylic Acetals

Maddaluno and collaborators of the IRCOF group in Rouen have published important papers on the use of organometallic polar reagents in organic synthesis. In particular, they have studied the reactivity of unsaturated acetals in conjugate elimination and addition processes, the role of chiral ligands in some stereoselective syntheses promoted by alkyllithium reagents, the nature and the role of mixed aggregates both from an experimental and computational point of view.

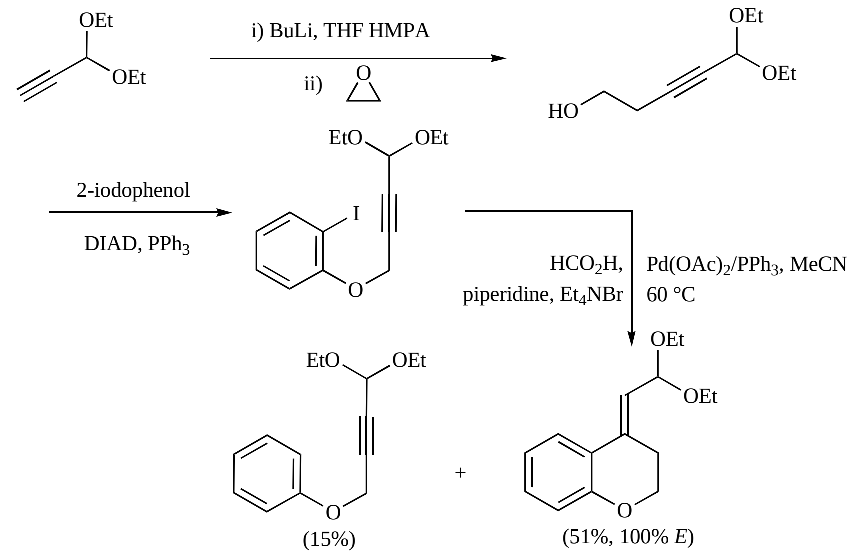
A new synthetic pathway affording binuclear vinyl heterocycles from propargylic acetals has been reported. In particular, 2-vinylbenzofurans, 3-vinylfuro-[2,3-c]pyridine, 4-vinylisochromenes, and 4-vinylisoquinolinones have been prepared. As an example, in Scheme **26** is shown the synthesis of 3-vinylfuro[2,3-c]pyridine **86**, starting from 4-iodo-3-hydroxypyridine **81** and propargylic bromide **82** [24]. On the other hand, when iodopyridyl ether **83** was treated with *tert*-BuLi, the cyclization product was found to be the minor product, with the deiodinated allene **84,** the major component of the mixture.

In following papers significant experimental and theoretical details have been added concerning the carbolithiation step and the unexpected sterochemical course that starting from (4,4-diethoxybut-2-ynyloxy)benzene pilots to 100% *E* 3-(2-ethoxyvinyl) benzofuran [25-26]. The NMR spectroscopic study on the adduct has shown that this reaction results from an unprecedented *anti* addition on the alkyne. In fact, DFT calculations indicate that the reaction proceeds through a low-lying transition state responsible for the *anti* addition to the alkyne, and that intramolecular coordination of lithium by one oxygen atom of the acetal group is responsible for this stereochemical outcome. The O–Li interaction is observed throughout the cyclization and drives the cation to the appropriate site on the double bond, finally leading to the formation of the *E* product. The calculations show moreover, that in absence of this coordination the *Z* olefin that would result from a classical *syn* addition should be obtained.

A putative mechanism for this anionic heterocyclization of propargylic **87** and allenic acetal **88** was also proposed (Scheme **27**).



Scheme 27. Synthesis of vinyl heteronuclear bisystems.



Scheme 28. Synthesis of [4.4.0] systems by a Palladium induced cyclization.

The procedure results a useful entry to a broad range of vinyl-[4.3.0] binuclear heterocycles but is of little value in the synthesis of homologous [4.4.0] systems. The latter are readily prepared by means of a two-step procedure involving a palladium-induced cyclization as reported in Scheme **28**.

1,3-DIENES

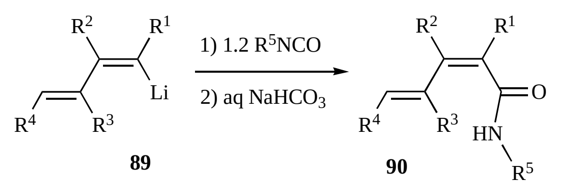
1-Lithium-1,3-Dienes

The alkenyllithium moiety of 1-lithio-1,3-butadienes reacts with a variety of organic substrates, with a pathway influenced by the nature of the substituents. The intramolecular synergy between the alkenyllithium moiety and the butadienyl skeleton makes 1-lithio-1,3-butadienes derivatives useful buiding blocks in organic synthesis.

1-Lithio-1,3-butadienes are prepared generally *in situ* by lithium-halogen exchange of the corresponding 1-halo-1,3-butadienes, or mixed 1,4-halo-1,3-butadienes, with BuLi, or *t*-BuLi.

Xi and Zhang developed new reactions and applications based upon monolithio reagents, providing buta-1,3-dienyl-containing amides, alcohols, imines, and related compounds [27].

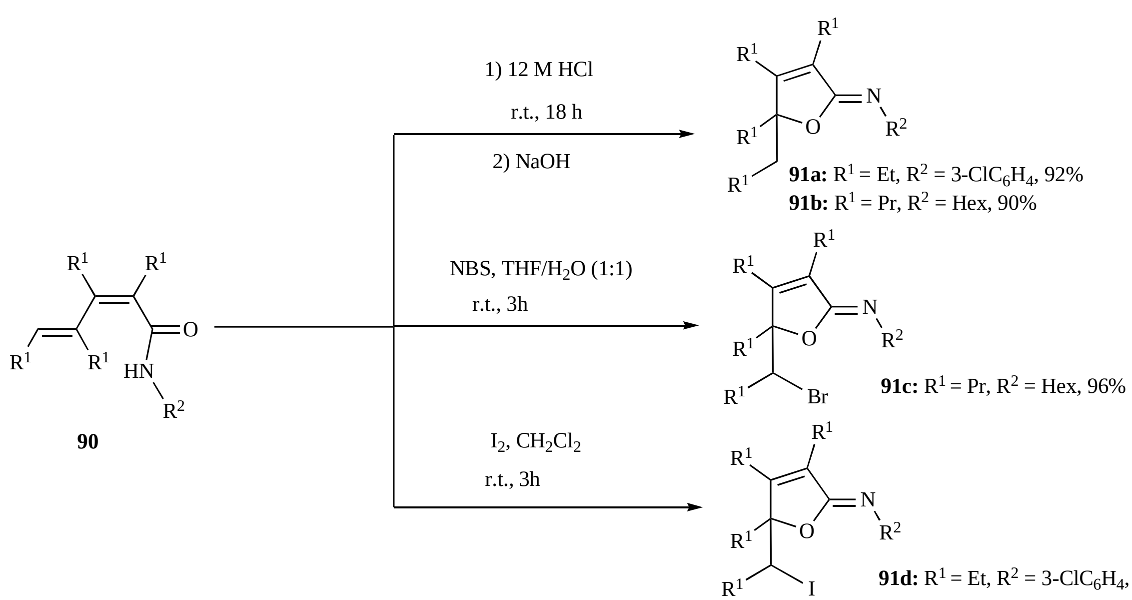
In particular, conjugated dienamides **90** were synthesized by the reaction of the corresponding 1-lithio-1,3-dienes **89** with aryl or alkyl isocyanates (Scheme **29**).



89 **90**

Scheme 29. Formation of dienylamides by the reaction of butadienyl-lithium reagents with isocyanates.

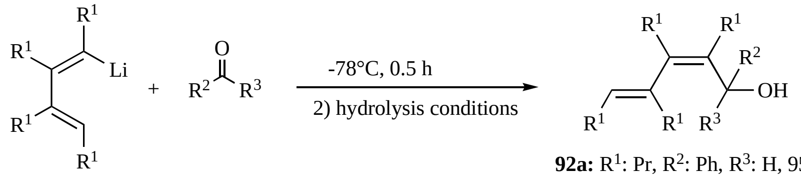
Exocyclic imino ethers **91** were selectively obtained by the electrophilic cyclization promoted by an acid, *N*-bromosuccinimide, or iodine, through the O-attack pathway (Scheme **30**).



90

91a: R^1 = Et, R^2 = 3-ClC_6H_4, 92%
91b: R^1 = Pr, R^2 = Hex, 90%
91c: R^1 = Pr, R^2 = Hex, 96%
91d: R^1 = Et, R^2 = 3-ClC_6H_4, 95%

Scheme 30. Formation of exocyclic imino ethers through cyclization of multisubstituted dienamides.



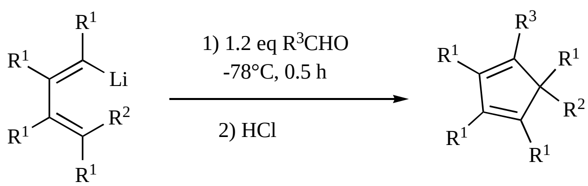
89

92a: R^1: Pr, R^2: Ph, R^3: H, 95%
92b: R^1: Pr, R^2: Pr, R^3: H, 82%
92c: R^1: Pr, R^2: Pr, R^3: Pr, 71%
92d: R^1: Et, R^2: PhC_6H_4, R^3: H, 72%

Scheme 31. Formation of conjugated dienols by the reaction of 1-lithiobuta-1,3-dienes with aldehydes or ketones.

The same work reported the selective synthesis of conjugated dienols **92** by the hydrolysis of the reaction mixture of 1-lithiobutadienes **89** and aldehydes or ketones (Scheme **31**).

In an another work [28], Xi and co-workers reported that the conjugated dienols obtained by 1-lithio-1,3-butadiene derivatives substituted with a methyl, a phenyl group, or a hydrogen at position 4 of the butadienyl skeleton, were transformed into their cyclopentadiene derivatives **93** when treated with a strong acidic solution (Scheme **32**). In the presence of a methyl or a phenyl substituent at position 4, stronger acid (12N HCl) was required for the formation of cyclopentadiene derivatives, while without substituent at that position, weaker acid (3N HCl) could promote the cyclization.



R^2 = Me, Ph, H

89

93

Scheme 32. Formation of cyclopentadiene derivatives.

When (1-lithiobuta-1,3-dienyl)diphenylphosphine oxides **94** reacted with various aldehydes, stereodefined vinylallenes **95** were formed by the potassium *t*-butoxide promoted Wittig-Horner reaction, while the dienols **96** (Scheme **33**) were formed through the hydrolysis of the intermediate lithium alkoxides.

As described in Scheme **34**, linear butadienyl-containing imines **97** were obtained by the reaction of 1,2-disubstituted 1-lithiobuta-1,3-dienes with organonitriles at –78°C, when the temperatures was increased, pyrrole derivatives **98a** appear in addition to linear imines, and if the reaction was carried out at reflux temperature, pyrrole derivatives were obtained as the sole products. In the case of PyCN, the bipyridine derivative was obtained as a consequence of the coordination of the nitrogen atom to the Li atom (intermediate **99**) [29].

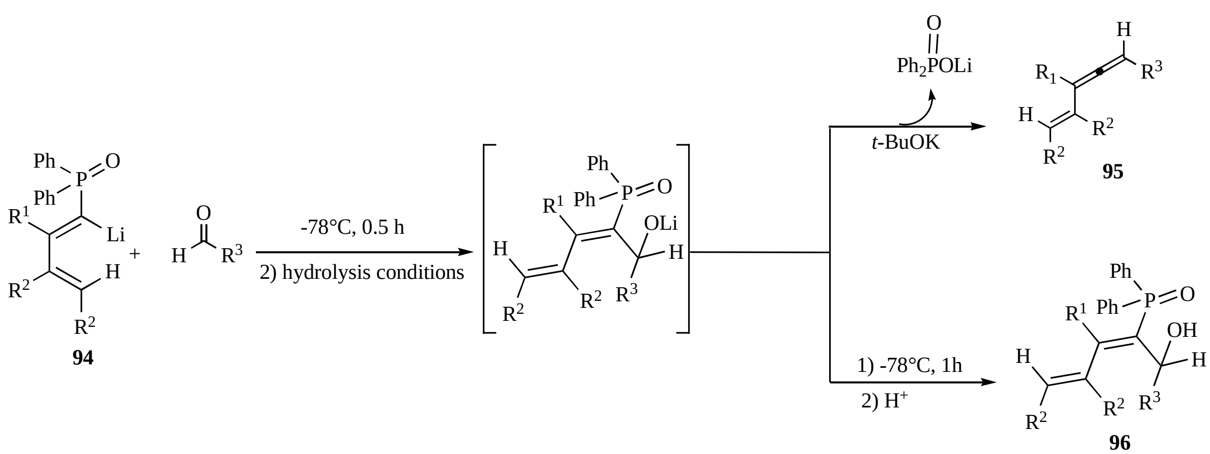
Generally speaking, the substituents on the butadienyl skeleton remarkably influenced the substitution pattern; in fact, as summarized in Scheme **35**, *N*-lithioketimines **100** obtained from 1,2,3,4-tetrasubstituted and from 2,3-disubstituted lithium reagents underwent 6-*endo* cyclization to give pyridine derivatives **101**, while those obtained from 1,2-disubstituted lithium reagents underwent 6-*endo* and 5-*exo* competitive cyclization to give either pyridine or pyrrole derivatives **102**, depending on the reaction conditions.

The same *N*-lithioketimine intermediate **100** was assumed to be formed also in the reaction of 1-cyano-1,3-butadiene **103** with PhLi (Scheme **36**), but in this case the pyridine derivative **101** was obtained in a low yield [30].

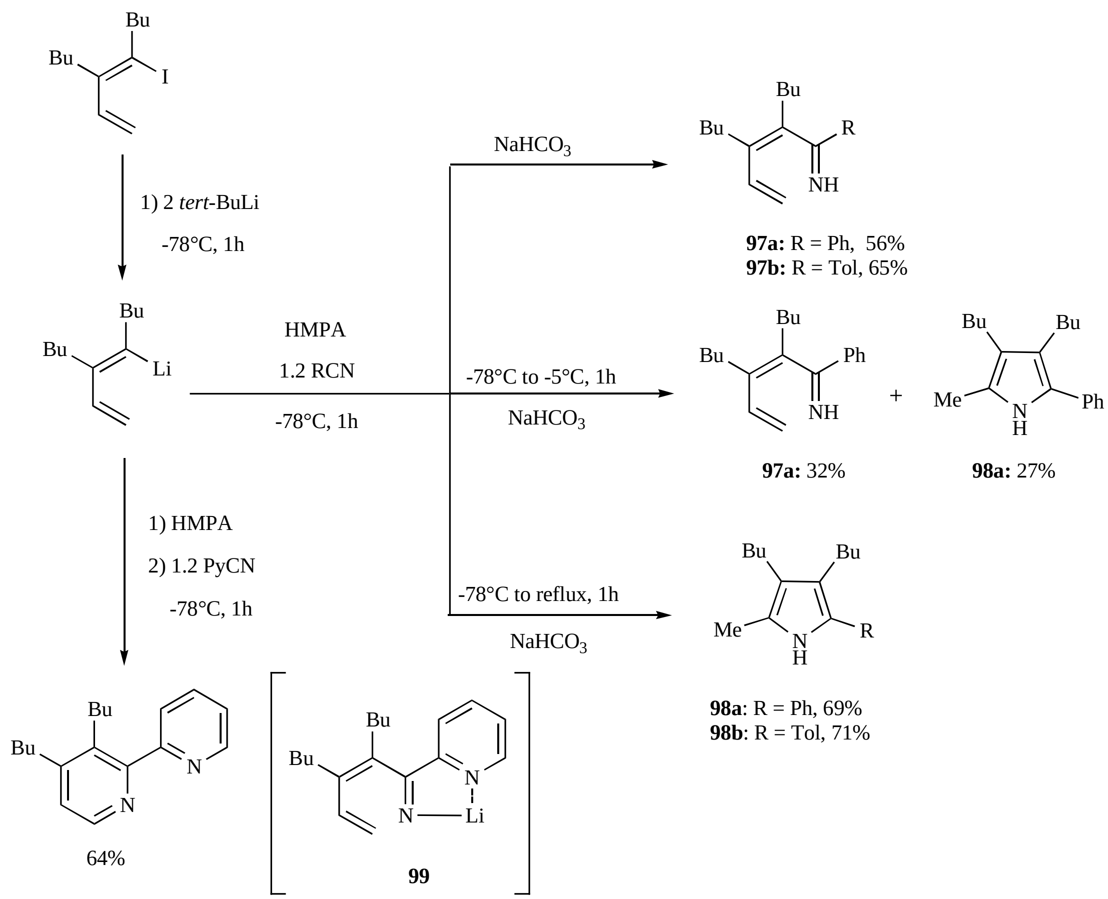
Surprisingly, 2*H*-pyrrole derivatives **105** were formed in good yields through the reaction of 4-halodienenitriles **104** and organolithium reagents. When the reaction was performed at -50°C for one hour, the linear imine **106** was obtained through the quenching of the reaction mixture (Scheme **37**).

In an another work [31], the same authors described the intramolecular nucleophilic addition of butadienyllithium to the aromatic rings of 4-phenyl-1-lithio-1,3-butadienes **108** (Scheme **38**) and 4-naphtyl-1-lithio-1,3-butadienes **111** (Scheme **39**), affording spirocyclopentadiene derivatives **109** and **110** as full or partial dearomatization products of aromatic rings.

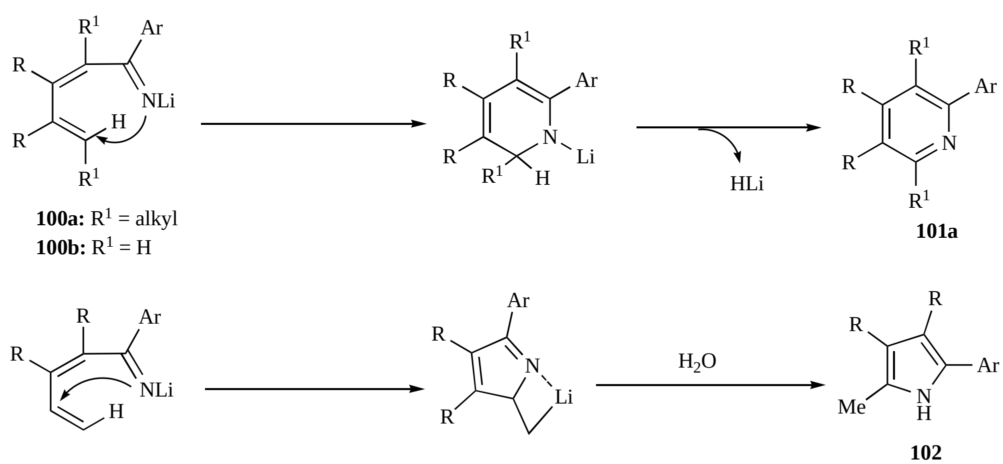
Maddaluno and co-workers have also presented a new synthetic pathway affording 1,2,4-trioxygenated 1,3-dienes **116**, exploiting a δ-elimination reaction on α,β-unsaturated acetals bearing a silyloxy



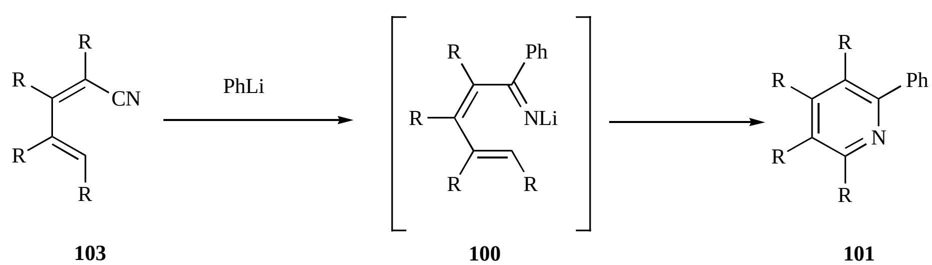
Scheme 33. Reactions of (1-lithiobuta-1,3-dienyl)phosphine oxides with aldehydes.



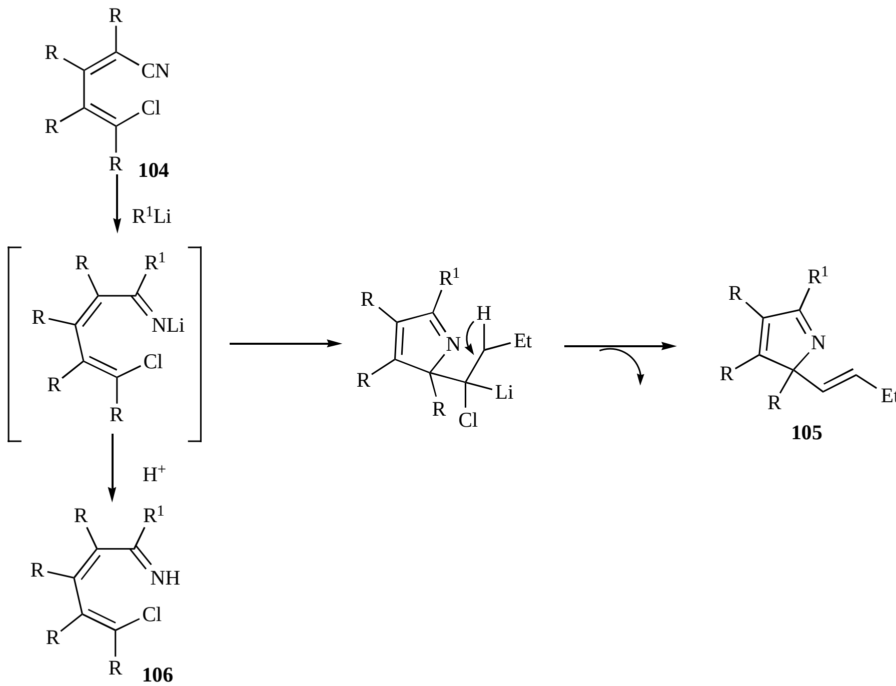
Scheme 34. Reaction of 1,2-disubstituted 1-lithiobuta-1,3-dienes with organonitriles.



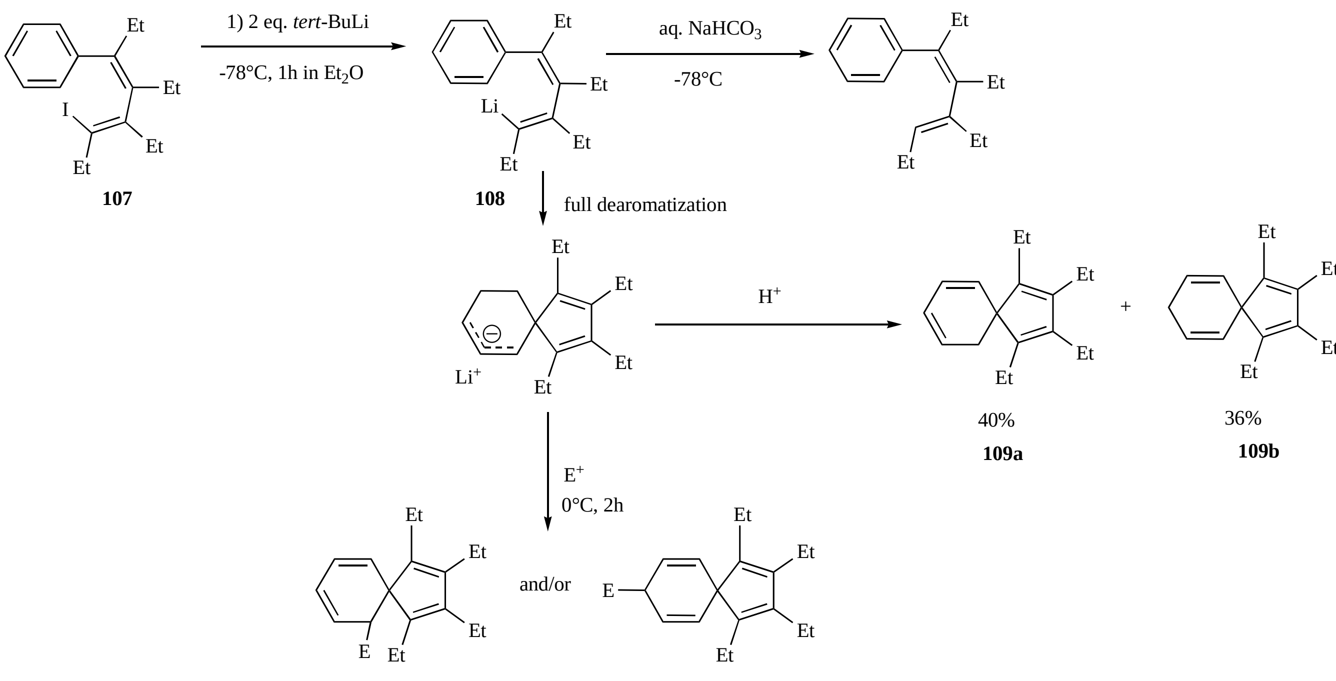
Scheme 35. Summary of the reactivity of cycloaddition intermediates of 1-lithio-1,3-dienes with organonitriles.



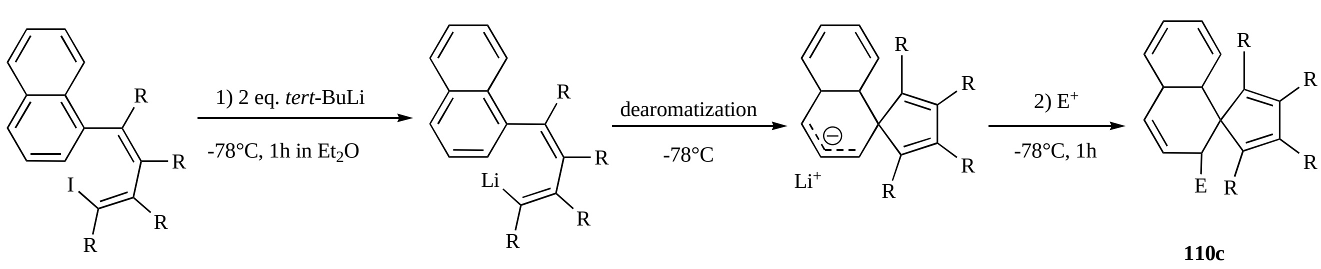
Scheme 36. Reaction of 1-cyano-1,3-butadiene with organolithium reagent.



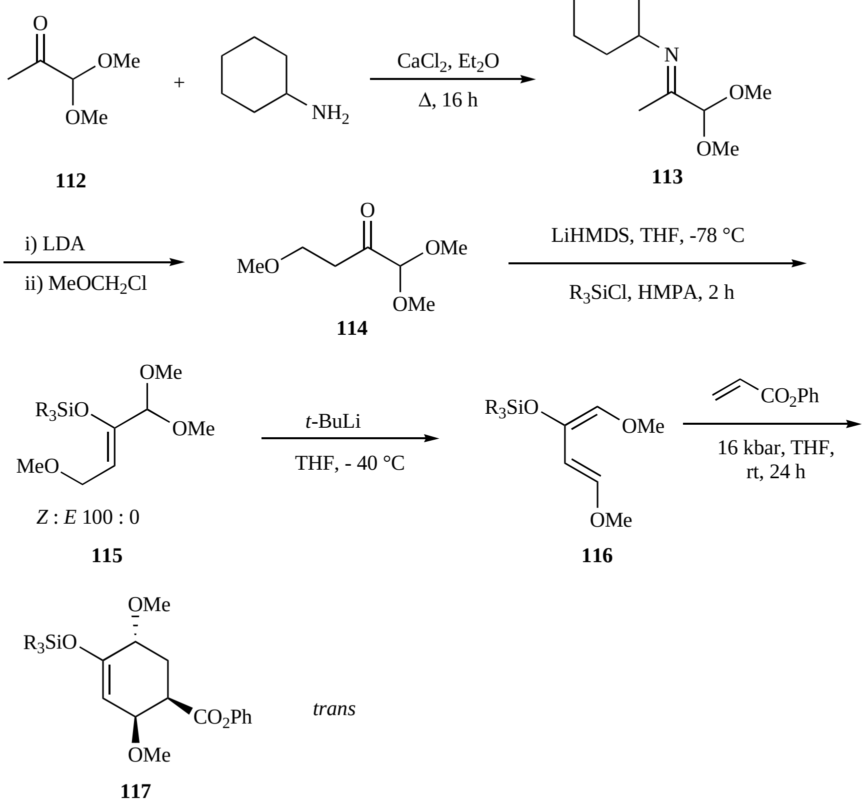
Scheme 37. Proposed mechanism for the reaction of 4-chlorodienenitriles and an organolithium reagent.



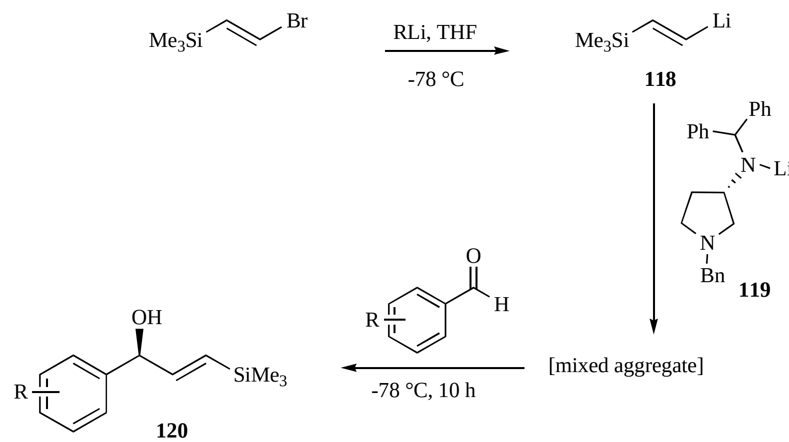
Scheme 38. Dearomatizing anionic cyclization of 1-lithio-4-phenyl-1,3-butadienes.



Scheme 39. Dearomatizing anionic cyclization of 1-lithio-4-naphtyl-1,3-butadienes.



Scheme 40. 1,2,4-trioxygenated 1,3-dienes.



Scheme 41. Use of chiral 3-aminopyrrolidine lithium amides in asymmetric nucleophilic vinylations of aromatic aldehyde.

group at C^2, triggered by *t*-butyllithium at low temperature (Scheme **40**). The starting 1,2,4-trioxygenated 1,3-dienes **116** have been prepared from commercially available pyruvic aldehyde dimethyl acetal **112** in four synthetic steps. Hyperbaric [4 + 2] cycloadditions between **116** and *N*-methylmaleimide or methyl- and phenylacrylates give access to the expected cycloadducts **117** with fine stereo- and regiocontrol [32].

These authors have also shown that bulky vinyllithium derivatives can be used as mixed aggregates with chiral 3-aminopyrrolidine lithium amides **119** in asymmetric nucleophilic vinylations of aromatic aldehydes in average yields but in ee up to 70% in standard conditions (–78 °C, THF), as shown in Scheme **41**, in which the reaction of (*E*)-2-trimethylsilylvinyllithium **118** is reported [33].

In some related studies it has been supported the idea that mixed aggregates can become useful tools in asymmetric synthesis. The tight non-covalent association of a chiral entity to a very reactive nucleophile can indeed constitute a perfect couple provided that the chelation and solvation phenomena are attuned to the task. Spectroscopic and theoretical studies help to understand the mechanisms underlying the induction phenomena. The formation of strong 1:1 noncovalent aggregates between the 3-aminopyrrolidine lithium amides and the alkyllithium derivatives could be established in THF and Et_2O. Actually, aryl- and vinyllithium behave similarly. Multinuclear NMR and quantum mechanics both suggest that these entities are built around a N–Li–C–Li quadrilateral, in which one of the two lithium cations is chelated by the nitrogen of the pyrrolidine ring. A folded norbornyle-like arrangement results from this intramolecular interaction. When the lateral chain borne by the exo-

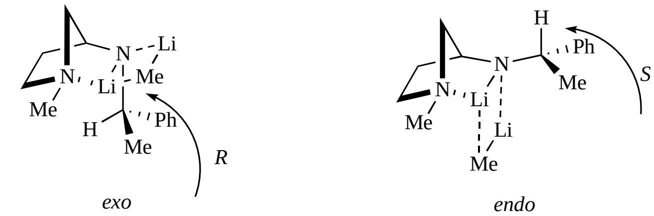
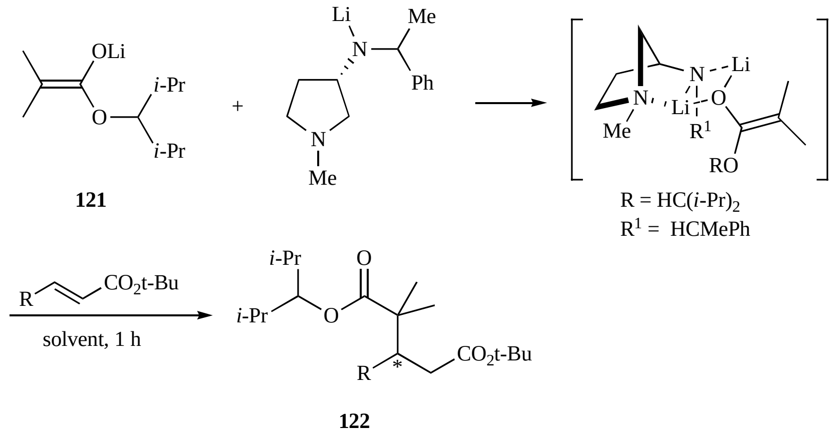
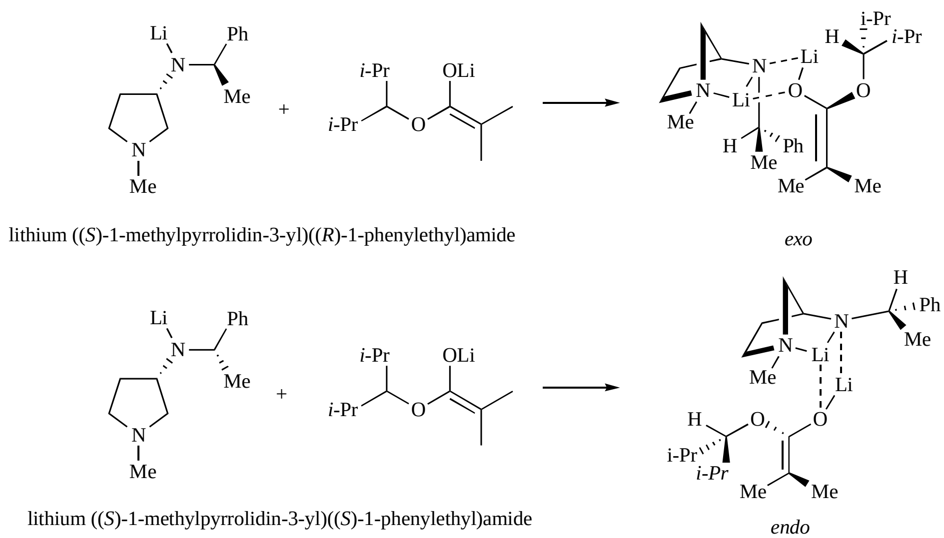


Fig. (2). *Endo* or *exo* topology of chiral 3-aminopyrrolidines.



Scheme 42. 3-Aminopyrrolidines as chiral auxiliaries in enantioselective Michael additions of lithium enolate to α,β-unsaturated esters.



Scheme 43. Mixed aggregates between a lithium enolate and two diastereomeric lithium amides.

cyclic nitrogen is chiral, an *endo* (concave) or *exo* (convex) topology can be obtained, depending on the relative configuration of the two asymmetric centers (Fig. **2**).

The sense of the asymmetric induction during the condensation of the aggregates on *o*-tolualdehyde seems to be controlled by these topologies, and thus by the configuration of the original amide.

In a subsequent study [34] these authors have reported similar results showing that 3-aminopyrrolidines can also form aggregates with lithium enolates of ester to be used as chiral auxiliaries in enantioselective Michael additions of lithium enolate **121** to α,β-unsaturated esters. Values for ee's of up to 76% were reached after optimization of the experimental conditions. These results demonstrate that the sense of the induction appears to be controlled both by the configuration of the stereogenic centers borne by the amide and by the solvent.

Two 1:1 noncovalent mixed aggregates between a lithium enolate and two diastereomeric lithium amides have been identified spectroscopically in THF, and described in a recent paper [35] (Scheme **43**).

In a recent investigation on the enantioselective nucleophilic addition of methyllithium on aryl aldehydes, in the presence of chiral lithium amides derived from chiral 3-aminopyrrolidines Maddaluno and colleagues have found that the enantiomeric excess of the resulting 1-arylethanol goes down upon addition of significant amounts of lithium halides, added to the reaction mixture before the aldehyde [36]. The results show that the original aggregate of the chiral lithium amide and methyllithium is replaced by a similar 1:1 complex involving one lithium halide and one lithium amide. While the MeLi/LiX substitution occurs with some degree of epimerization at the nitrogen for the *endo*-MeLi: 3-

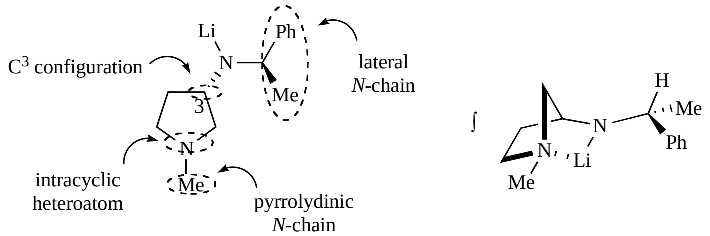
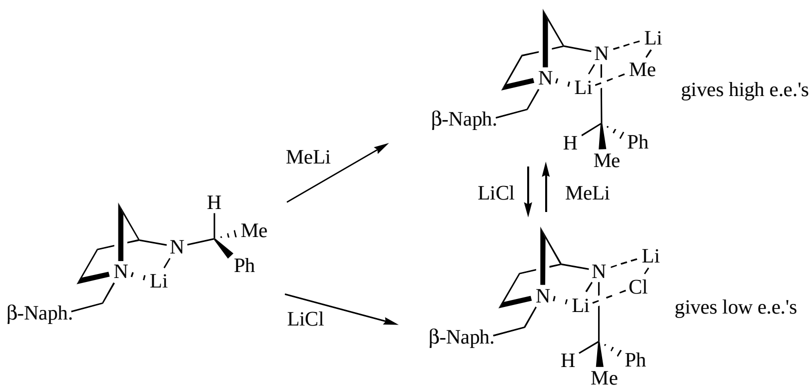


Fig. (3). 3-Aminopyrrolidine lithium amides.



Scheme 44. *Exo*-type arrangements of lithium amides.

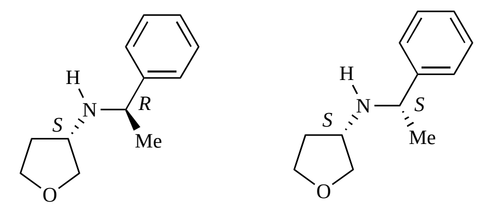
aminopyrrolidine lithium amides it is mostly stereospecific for the exo-type arrangements of the aggregate (Scheme **44**).

The formation of the various possible aggregates between the LiX, the lithium amide, and the methyllithium was examined by multinuclear NMR spectroscopy in THF. DFT calculations confirm the thermodynamic preference for the 3-aminopyrrolidine lithium amides: LiX mixed aggregates (by more than 10 kcalmol^{-1}).

As an attractive extension of the above findings, Maddaluno and other have recently published a paper that presents a general synthetic pathway to synthesize easily new chiral structures based on 3-aminopyrrolidine, 3-aminotetrahydrothiophen, and 3-aminotetrahydrofuran scaffolds [37]. Their lithium derivatives have been evaluated as chiral ligands in the condensation of butyllithium on *o*-tolualdehyde. The 1-*o*-tolylpentan-1-ol was isolated in good yields and ee's up to 80% in favor of the *R* or *S* enantiomer, depending on the absolute configuration of the lateral chiral amino appendage.

In particular, as shown in Fig. (**3**), four structural appendages on 3-aminopyrrolidine lithium amide have been varied, namely intracyclic heteroatom, pyrrolydinic *N*-chain, C^3 configuration, and lateral *N*-chain.

These attractive values were obtained using the simple, cheap and easy to prepare lithium amides obtained by treating with BuLi the (*S*)-*N*-((*R*)-1-phenylethyl)tetrahydrofuran-3-amine **123**, or the



(*S*)-*N*-((*R*)-1-phenylethyl) tetrahydrofuran-3-amine

123

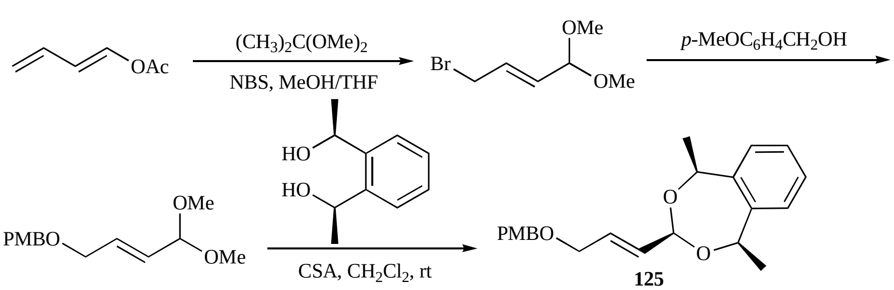
(*S*)-*N*-((*S*)-1-phenylethyl) tetrahydrofuran-3-amine

124

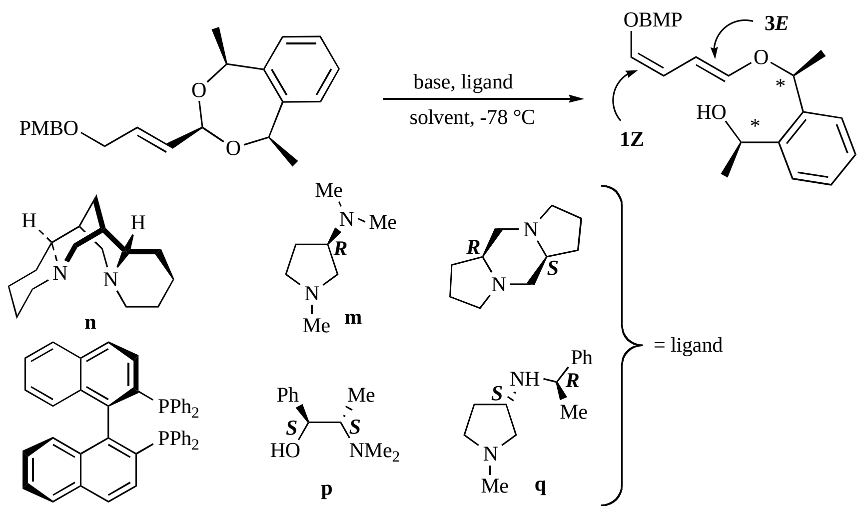
Fig. (4). (*S*)-*N*-((*R*)-1-phenylethyl)tetrahydrofuran-3-amine and (*S*)-*N*-((*R*)-1-phenylethyl)-tetrahydro-furan-3-amine.

(*S*)-*N*-((*R*)-1-phenylethyl)-tetrahydrofuran-3-amine diasteroisomer **124** shown in Fig. (**4**).

Maddaluno and collaborators have recently considered that while the asymmetric conjugate addition of organometallic nucleophiles on activated olefins has been the object of extensive attention also by other research's groups, the conjugate base-triggered elimination has drawn much less consideration [38]. Moreover, no asymmetric version has been previously proposed. Thus, in order to develop an enantioselective version of conjugate elimination these authors have thought that the desymmetrization of an unsaturated *meso*-1,3-dioxepane by a chiral base could solve the problem. The suitable cyclic acetal **125** was synthesized according to Scheme **45**,



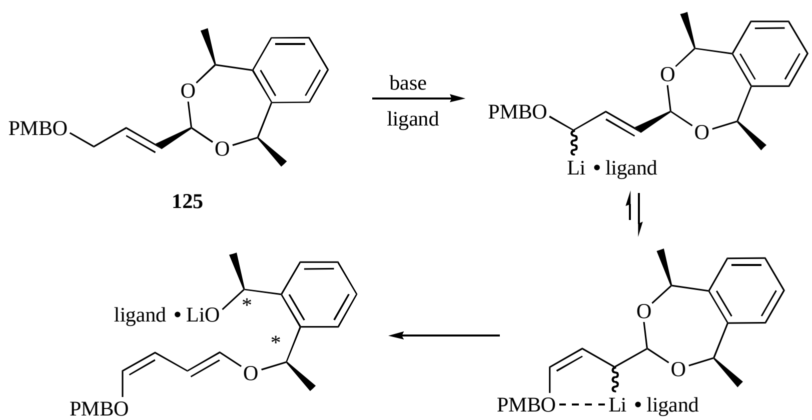
Scheme 45. Synthesis of α,β–unsaturated acetals.



Scheme 46. Enantioselective deprotonation/acetal opening sequence in the presence of chiral ligands.

Table 1. Yields and ee's of the Enantioselective Conjugate Elimination of 125 Under the Action of Various Bases and Ligands

Base	Ligand	Solvent	Yield (%)	ee (%)
sec-BuLi	-	THF	78	-
tert-BuLi	-	THF	61	-
LDA	-	THF	60	-
sec-BuLi	n	THF	81	30 (–)
sec-BuLi	n	Et_2O	72	64 (–)
sec-BuLi	n	DMM	27	70 (–)
sec-BuLi	m	Et_2O	34	10 (–)
sec-BuLi	p	toluene	31	27 (+)
sec-BuLi	q	DMM	91	15 (+)



Scheme 47. Deracemization of α,β-unsaturated acetals.

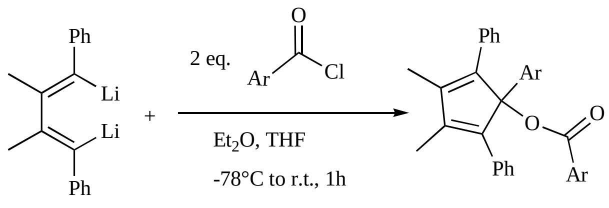
and the enantioselective deprotonation/acetal opening sequence was then studied varying the nature of the base (*sec*-BuLi, *tert*-BuLi, LDA), the solvent (THF, Et_2O, DMM, toluene), and the chiral ligand, as shown in Scheme **46**. Same results are reported in Table **1**.

n the one hand, at this stage the authors have not yet totally cleared the details of the mechanism of the conjugate elimination, and therefore the induction process occurring in the enantioselective process that they have described. On the other hand, they correctly underline that actually, the two oxygens of the acetals are enantiotopic, and a specific interaction between the ligand-chelated lithium and one of the two oxygens is probably at the origin of the enantioselective process. The deracemization can result either at the deprotonation or at the elimination level. In this last case, a possible mechanism for the conjugate elimination has been suggested, and is shown in Scheme **47**.

1,4-Dilithio-1,3-Dienes

The most common method for the preparation of 1,4-dilithio-1,3-dienes is based upon the lithiation of the corresponding diiodo or dibromo compounds with *tert*- or straight-chain butyllithium at –78°C in 1 hour.

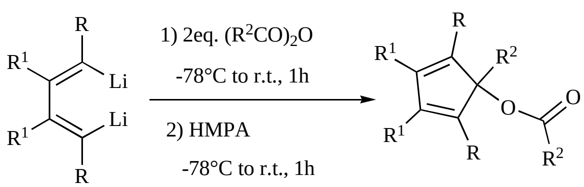
Most of the recent studies regarding synthetic applications of 1,4-dilithio-1,3-dienes belong to Xi and co-workers; for example, in 2007 they reported the synthesis of multiply substituted ciclopentadienes **128** through reactions between phenyl-substituted 1,4-dilithio-1,3-dienes **127** and acyl chlorides (Scheme **48**) [39].



127 **128**

Scheme 48. Reactions between 1,4-dilithio 1,3-dienes and acyl chlorides.

When treated with anhydrides such as $(PhCO)_2O$, and $(PrCO)_2O$, dilithiodienes furnished analogous type of multiply substituted cyclopentadienes **128** in excellent yields (Scheme **49**).



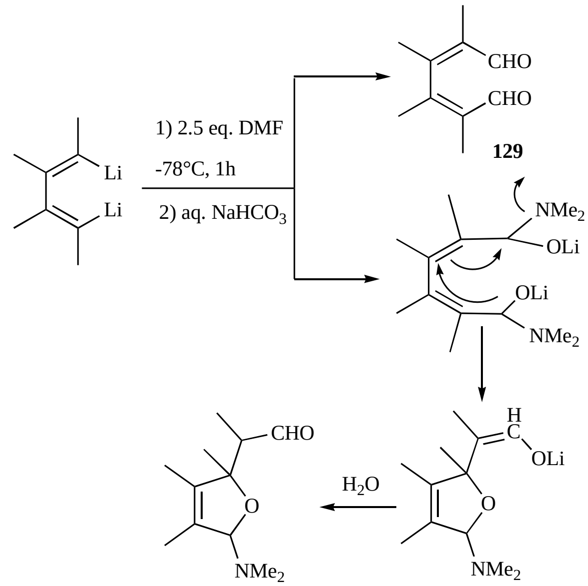
127a: R = R^1 = Pr
127b: R = $SiMe_3$, R^1 = Ph

128

Scheme 49. Reactions between 1,4-dilthio-1,3-dienes and anhydrides.

Diene-dials **129** were obtained from the reactions of 1,4-dilithio-1,3-dienes and DMF, in addition to the unexpected 2,5-dihydrofuran derivatives **130**, according to the mechanism reported in Scheme **50**.

In 2006 Xi *et al.* described the synthesis of multi-substituted semibullvalenes, starting from 1,2,3,4-tetrasubstituted-1,4-dilithio-1,3-diene, which was generated *in situ* from the corresponding 1,4-diiodo compound [40]. The semibullvalenes **131**, generated by the CuCl mediated dimerization of 1,4-dilithiodienes, could be thermally transformed to their corresponding cyclooctatetraene derivatives **132** (Scheme **51**).



129

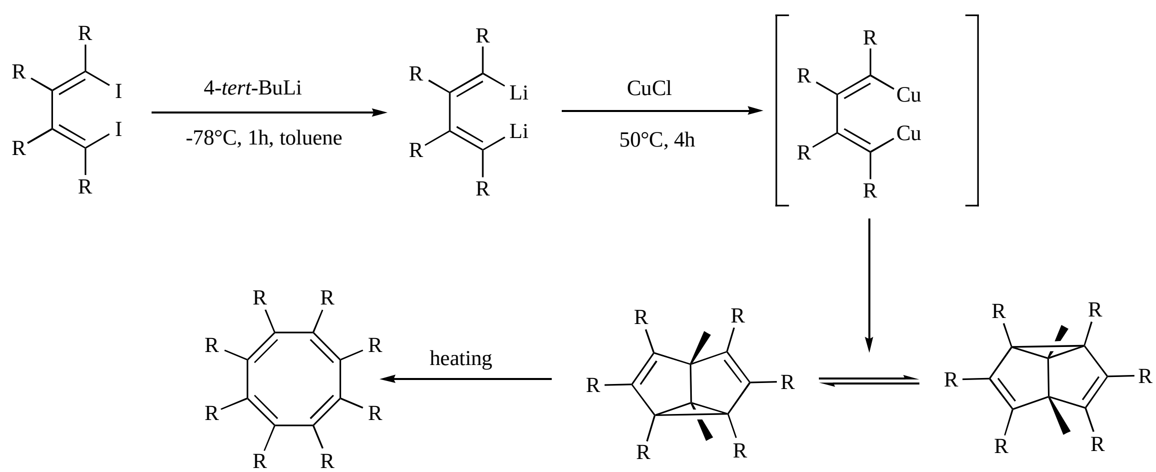
130

Scheme 50. Reaction of 1,4-dilithio-1,3-dienes with DMF.

In another work [41], the authors reported the formation and isolation of a new type of stable enols **134** from the reaction of 1,4-dilithio-1,3-butadienes **133** with acid chlorides, based upon a novel de-aromatization/Michael addition/rearomatization domino process (Scheme **52**).

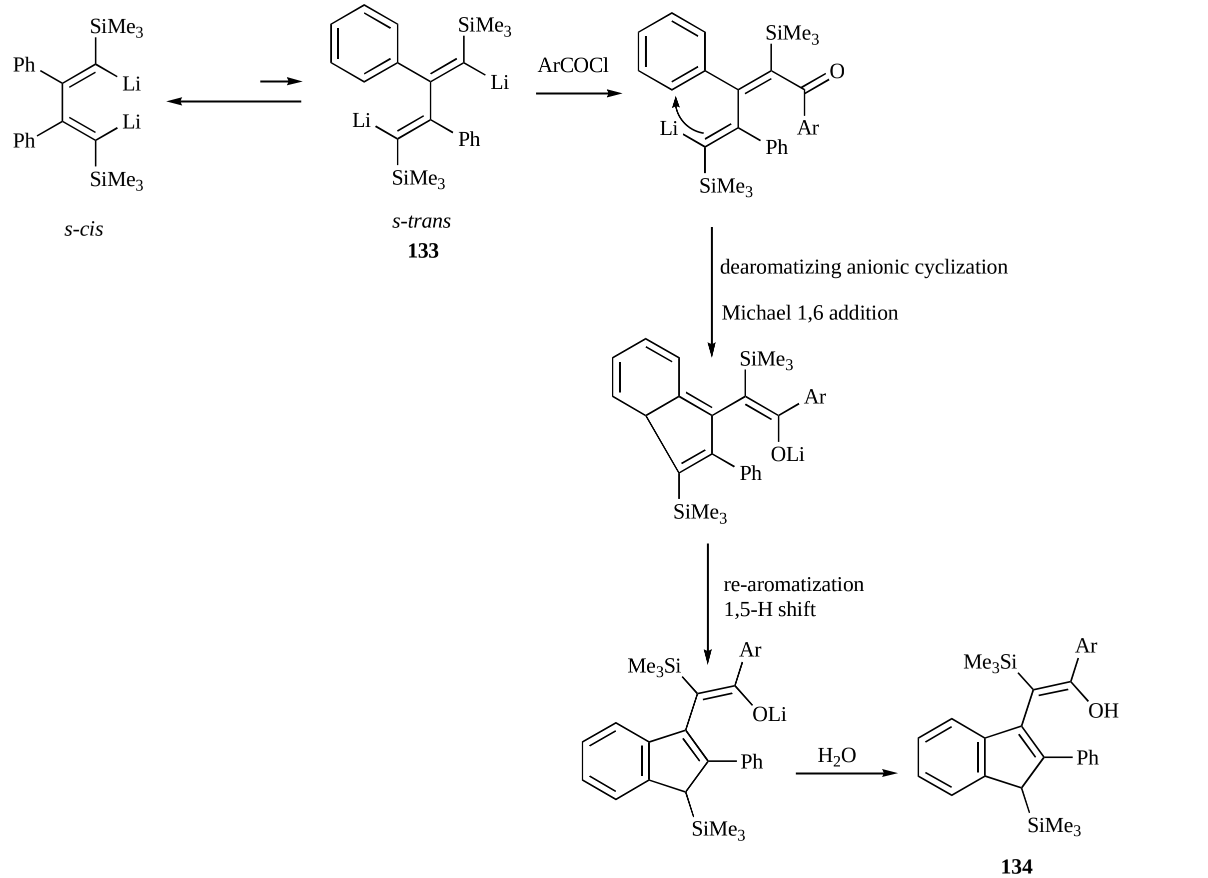
In 2007 the synthesis and applications of lithio siloles was described, starting from readily available silyl 1,4-dilithio 1,3-diene derivatives **133** which underwent novel intramolecular skeletal rearrangements, as shown in Scheme **53** [42]. The lithio siloles **134** thus obtained were variously substituted, giving access to various derivatives depending on the reaction conditions (**135** and **136**).

Xi and co-workers described also the CuCl mediated carbon monoxide insertion into 1,4-dilithio-1,3-dienes yielding multi-

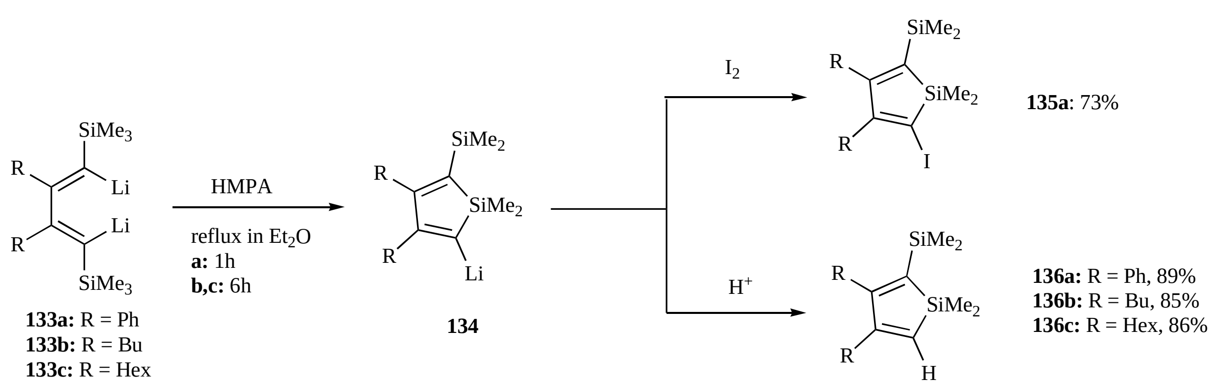


132 **131a** **131b**

Scheme 51. Synthesis of substituted semibullvalenes and cyclooctatetraenes.



Scheme 52. Proposed mechanism for the synthesis of the stable enols.



Scheme 53. Synthesis of functionalized siloles.

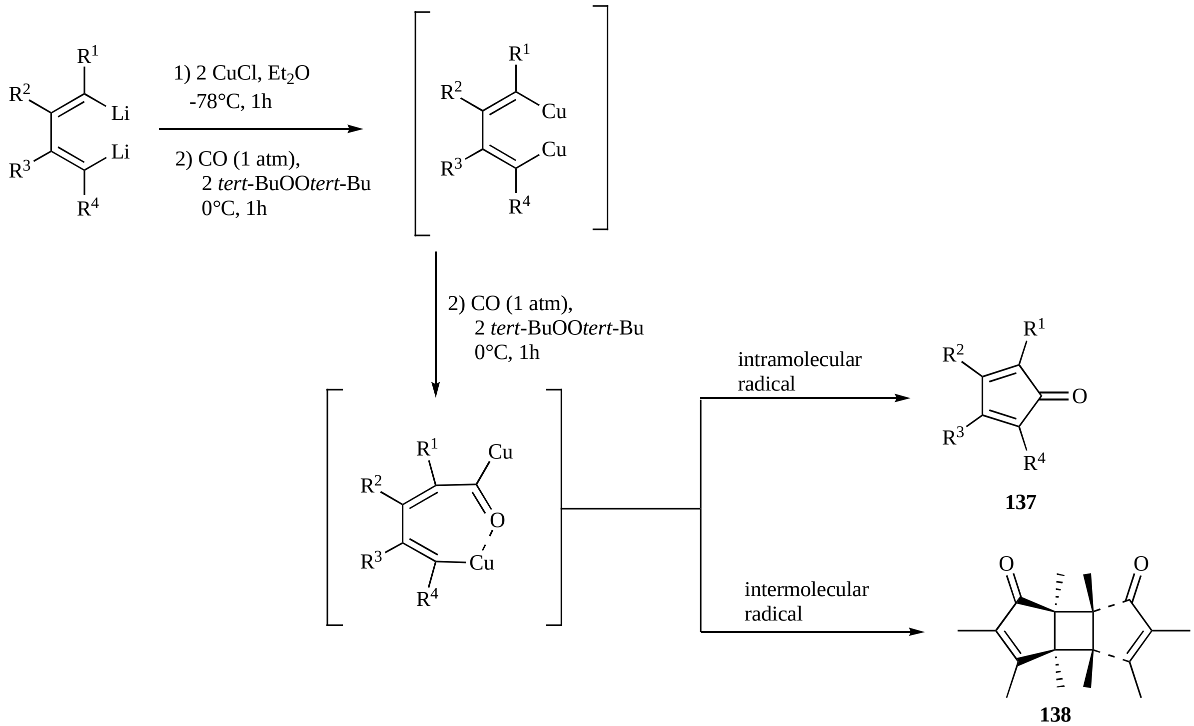
substituted cyclopentadienones **137** and/or their head-to-head dimers **138**, depending on the substitution pattern of dilithiobutadiene reagents [43]. The reaction proceeded *via* tandem CO insertion and intra/intermolecular cycloaddition of organocopper compounds (Scheme **54**).

The reaction between the linear dianion and CO afforded the fully substituted oxycyclopentadienyllithium dianions **139**, which showed novel reaction chemistry towards organic substrates and organometallic compounds [44]. For example, considering the reaction of **139** with acid chlorides, both the Cp ring and the oxy anion may be acylated (**141**), while when the reaction was carried out with two equivalents of acid chloride, doubly acylated cyclopentadienes **140** were obtained in good yields (Scheme **55**).

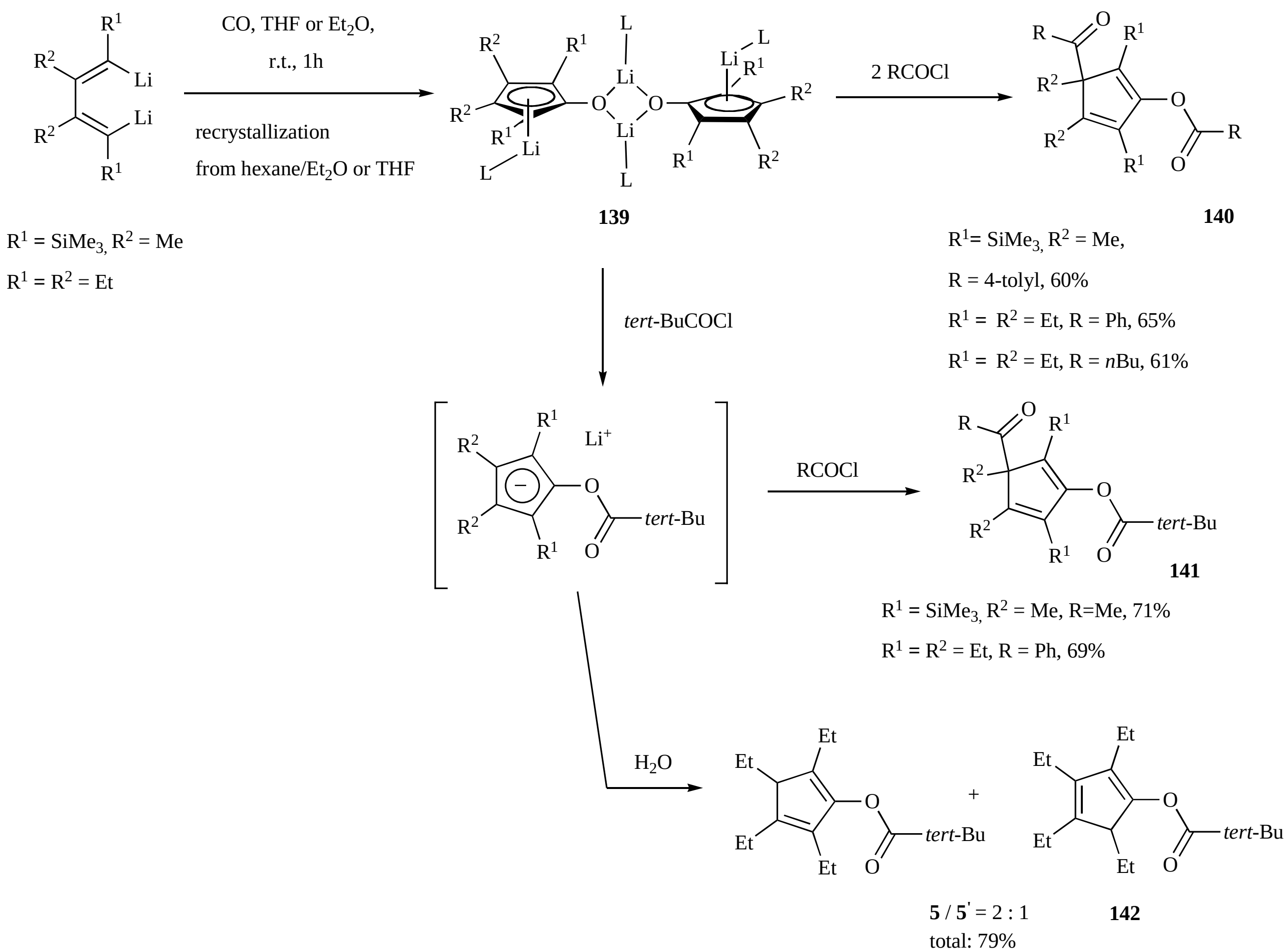
Saito and co-workers proposed a new way to synthesize encumbered 1,4-dilithio-1,3-butadiene derivatives reducing phenyl silyl acetylenes with lithium [45]. The reduction of phenyl(triisopropylsilyl)acetylene **143a** led to the unprecedent formation of the dilithium dibenzopentalenide **145** together with 1,4-dilithio-1,3-butadiene **144a**, whereas the reduction of phenyl(*tert*-butyldimethylsilyl)acetylene **143b** with lithium gave 1,4-dilithio-1,3-butadiene **144b** as the only product (Scheme **56**).

1-METALATED 1-ALKOXY-1,3-BUTADIENES

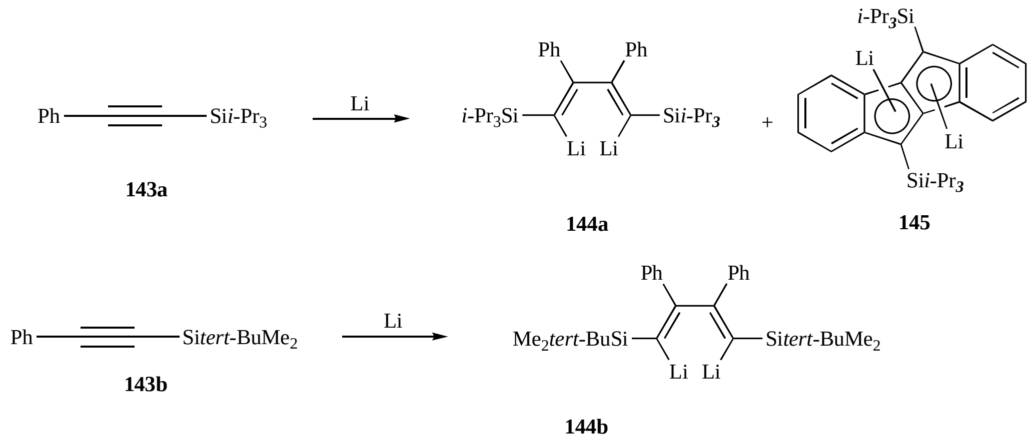
Since 1992 we studied the reactions of α,β-unsaturated acetals **145** with 2 equivalents of superbase LIC–KOR, affording 1-metalated-1-alkoxy-1,3-dienes [46]. According to an *umpolung*



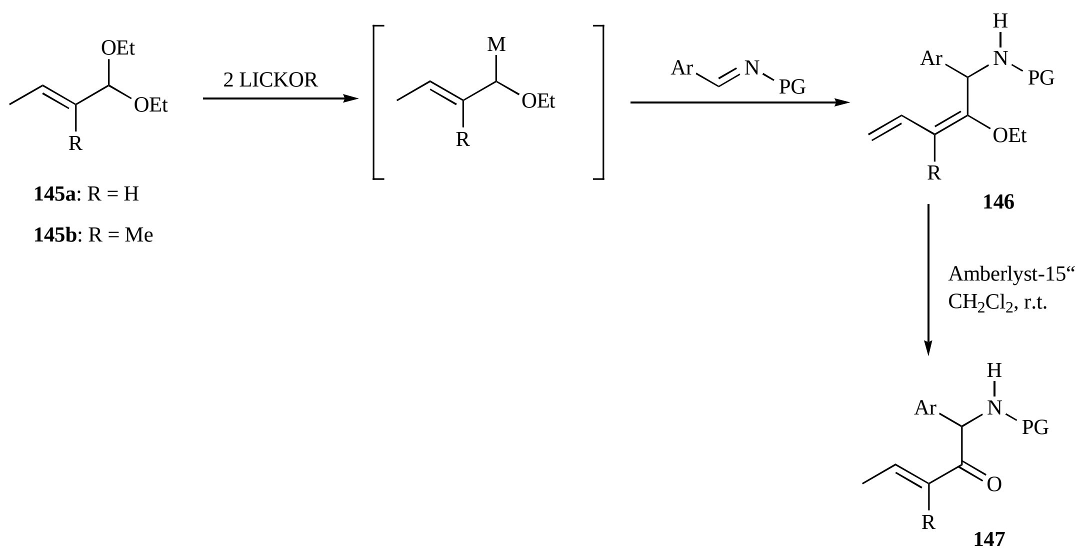
Scheme 54. Formation of cyclopentadienones and diketones.



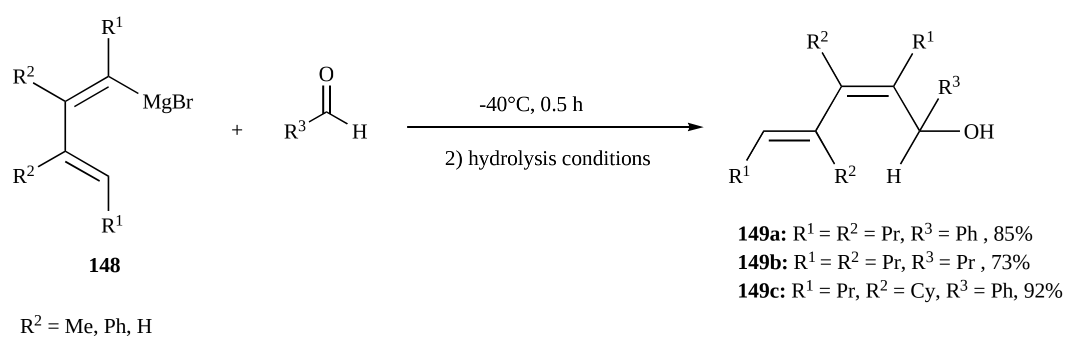
Scheme 55. Formation and reactivity of oxycyclopentadienyl dianions.



Scheme 56. Reduction of the phenyl silyl acetylene derivatives.



Scheme 57. Synthesis of alkoxydienyl amines and their use as α-aryl glycines precursors.



Scheme 58. Formation of conjugated dienols by the reaction of 1-(bromomagnesio)buta-1,3-dienes with aldehydes.

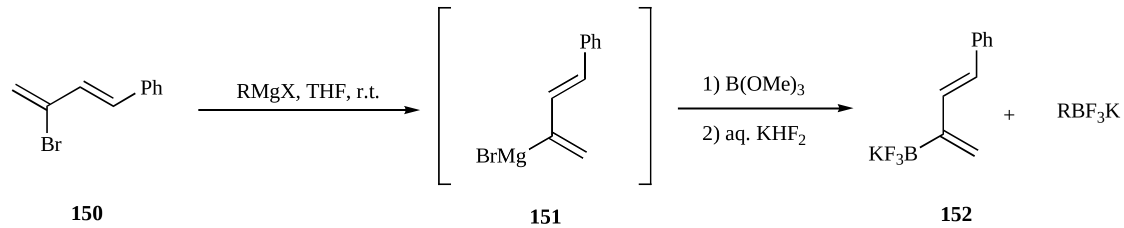
process, the metalated dienes thus obtained react with various electrophiles, such as alkyl halides, carbonyl compounds, and epoxides affording 1-alkoxyfunctionalized dienes.

During our recent studies regarding electrophiles containing multiple C-N bonds [47], we focused our attention on the use of imines as electrophiles. *N*-tosylaldimines bearing different aromatic rings, *N*-Tosyltolyl, *p*-methoxyphenyl, and thienyl, gave the best results, affording the corresponding alkoxydienyl amines **146** in good yields, which furnished *N*-protected α-arylglycines derivatives **147** when treated with Amberlyst (Scheme **57**) [48].

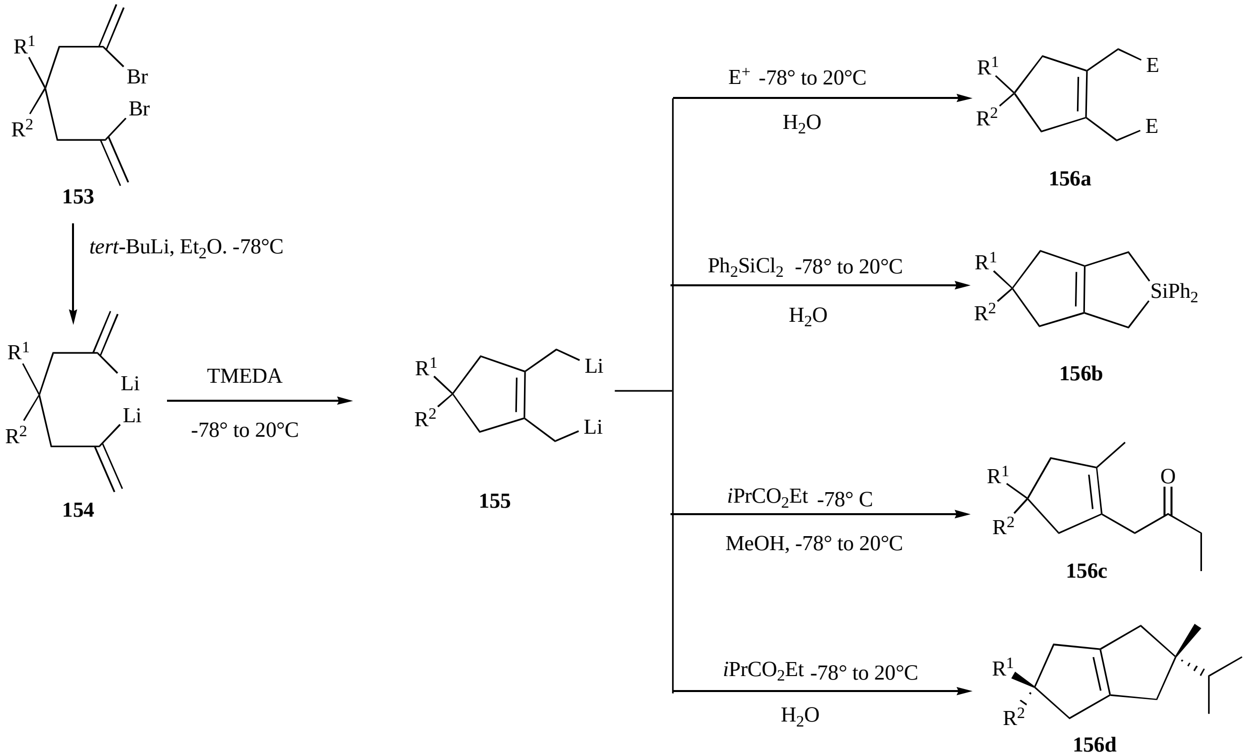
BROMOMAGNESIOBUTADIENES

In the work described above,[27] Xi and Zhang described also the formation of conjugated dienols **149** by the reaction of 1-(bromomagnesio)buta1,3-dienes **148** with aldehydes (Scheme **58**). As this derivatives did not react with ketones, their reactivity is different from that of 1-lithiobutadienes.

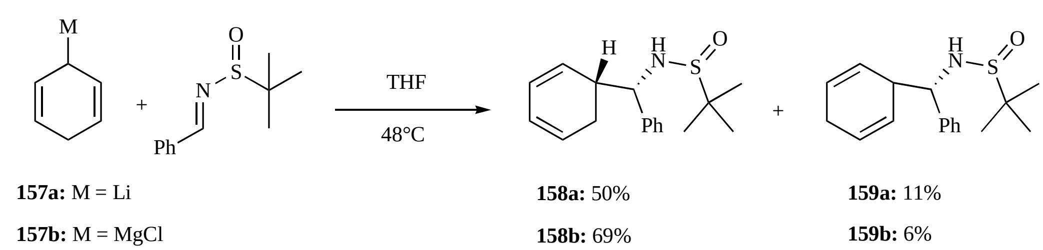
De, Day and Welker reported the synthesis of organomagnesium reagents **151** (obtained with halogen-magnesium exchange reactions), as useful intermediates to prepare 4-phenyl-2-BF_3-substituted 1,3-dienes **152** (Scheme **59**) [49].



Scheme 59. Halogen-magnesium exchange followed by boron electrophile.



Scheme 60. Intramolecular carbolithiation of 2,6-dilithio-1,6-heptadienes.



Scheme 61. Reaction of metallated cyclohexadienes with *N-tert*-butanesulfinyl imines.

MISCELLANEOUS

In 2007, Sanz and co-workers described intramolecular carbolithiation of 2,6-dilithio-1,6-heptadienes **154** affording 1,2-bis(lithiomethyl)cyclopentenes **155** [50]. Reaction of the dianions thus obtained with electrophiles afforded various 1,2-difunctionalized cyclopentene derivatives **156** (Scheme **60**).

In 2008, Maji, Fröhlich and Studer described the desymmetrization of various achiral metallated cyclohexadienes with chiral *N-tert*-butanesulfinyl imines, affording useful building blocks for natural product synthesis [51]. As shown in Scheme **61** the reactive cyclohexadienyl-Li compounds **157** provided the desymmetrization product **158a** along with an inseparable mixture of an isomer and the diene **159a**, while with the cyclohexadienyl-MgCl compound, the desymmetrized product **158b** was obtained with high diastereoselectivity (69% yield, ds > 99%) and the diene **159b** was the only side product.

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